Multivariable Calculus

MATH 1920 Lecture Notes

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1 Vectors in Two and Three Dimensions

Question: What information specifies a line?

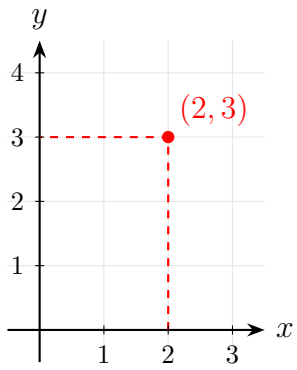
- 2 points
- A point and a direction vector
- Slope and y-intercept
- Intersection of 2 non-parallel planes

Comparison:

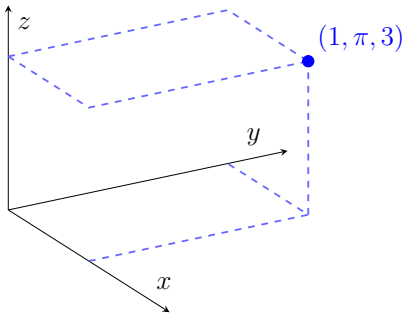
- **Single-variable calculus:** $f(x) \to$ rate of change of x
- **Multivariable calculus:** $f(x, y) \to$ functions of multiple variables

1.1 Coordinates

2-dimension (2D): $(2, 3)$



3-dimension (3D): $(1, \pi, 3)$



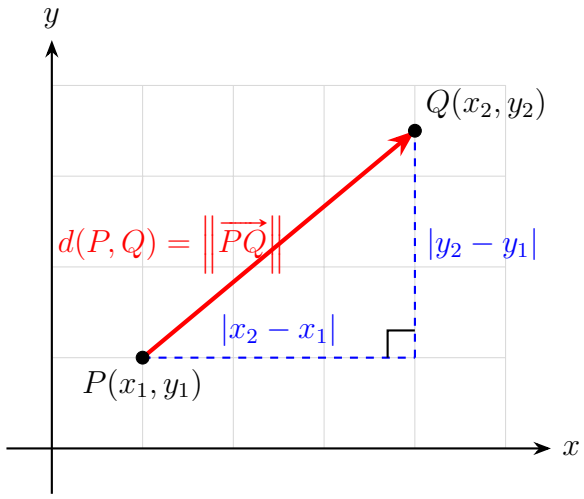
1.2 Distance Formulae

Definition 1.1: Distance Formula in $\mathbb{R}^2$

Let $(x, y) \in \mathbb{R}^2$ and points $P = (x_1, y_1)$, $Q = (x_2, y_2)$. The distance between P and Q is given by:

$$d(P, Q) = \left\| \overrightarrow{PQ} \right\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

This distance formula is a direct application of the Pythagorean theorem, where the distance $d(P, Q)$ forms the hypotenuse of a right-angled triangle with legs $|x_2 - x_1|$ and $|y_2 - y_1|$, as shown in the figure below.



The case in $\mathbb{R}^3$ follows similarly:

Definition 1.2: Distance Formula in $\mathbb{R}^3$

Let $(x, y, z) \in \mathbb{R}^3$ and points $P = (x_1, y_1, z_1)$, $Q = (x_2, y_2, z_2)$. The distance between P and Q is given by:

$$d(P, Q) = \left\|\overrightarrow{PQ}\right\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

1.3 Equations of Geometric Shapes

Consider the equation $x^2 + y^2 = 4$.

In $\mathbb{R}^2$, this equation is equivalent to the set:

$$\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 4\}$$

This represents a circle with radius $r = 2$ centered at the origin. The distance from the origin $(0, 0)$ to any point (x, y) on this circle is constant and equal to 2, which can be verified

directly using the distance formula:

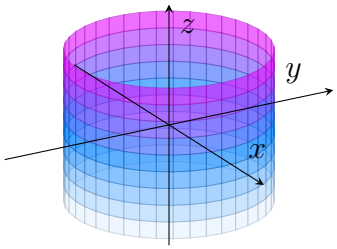
$$\sqrt{(x-0)^2+(y-0)^2}=2$$

Hence, the geometry of the circle is directly implied by the Pythagorean theorem.

Now consider the equation $x^2+y^2=1$ in three-dimensional space, $\mathbb{R}^3$. This set is formally expressed as:

$$\{(x,y,z)\in\mathbb{R}^3 : x^2+y^2=1\}$$

Because z is completely unrestricted, the cross-section at any constant height $z=c$ forms the unit circle $x^2+y^2=1$. Consequently, this equation describes a *right circular cylinder* extending infinitely along the z-axis, as shown.



1.4 Linear Equations across Dimensions

In $\mathbb{R}^2$, the general form of a linear equation is given by:

$$Ax+By+C=0 \quad (A,B,C\in\mathbb{R},\ \text{with } A \text{ and } B \text{ not both zero})$$

This equation represents a *line* in two-dimensional space.

In $\mathbb{R}^3$, however, the exact same algebraic equation $Ax+By+C=0$ represents a *plane* parallel to the z-axis (since z is a free variable), not a line.

To uniquely define a line in three-dimensional space, a single linear equation is insufficient; we require a system of *two* independent linear equations (representing the intersection of two non-parallel planes).

1.5 Vectors and Position Vectors

Definition 1.3: Equivalent Vectors

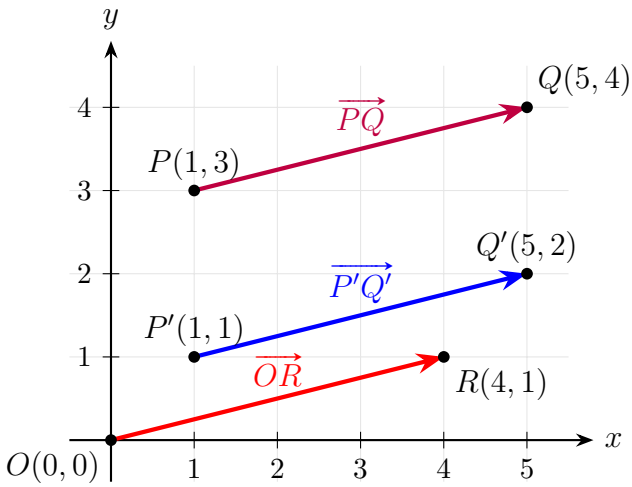
Two vectors are **equivalent** if one is a translation (pure shift without rotation) of the other.

Example 1.1.

Given points $P = (1, 3)$, $Q = (5, 4)$ and $P' = (1, 1)$, $Q' = (5, 2)$. Let $R = (4, 1)$ and $O = (0, 0)$. Then:

$$\overrightarrow{PQ} = \overrightarrow{P'Q'} = \overrightarrow{OR}$$

This could be indicated by the diagram below:



Definition 1.4: Position Vector

A vector whose initial point is at the origin $O(0, 0)$ is called a **position vector**. It is written using angle brackets:

$$\overrightarrow{OR} = \langle 4, 1 \rangle$$

Definition 1.5: Magnitude

The **length** or **magnitude** of a vector $\vec{v}$, denoted by $\|\vec{v}\|$, is the length of the line segment between its endpoints.

If $\vec{v} = \langle v_1, v_2 \rangle$, then its magnitude is given by:

$$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2}$$

For example, using $\overrightarrow{PQ} = \langle 4, 1 \rangle$:

$$\left\|\overrightarrow{PQ}\right\| = \sqrt{4^2 + 1^2} = \sqrt{17}$$

1.6 Unit Vectors

Definition 1.6: Unit Vector

A vector $\vec{v}$ is called a **unit vector** if its magnitude is 1, i.e., $\|\vec{v}\| = 1$.

In general, a unit vector of any function is given by:

$$\hat{v} = \frac{\vec{v}}{\|\vec{v}\|}$$

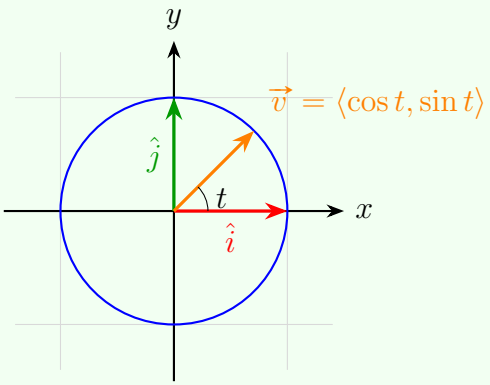
Problem 1.1: Describing Unit Vectors

Describe all unit vectors in $\mathbb{R}^2$.

Solution 1.1

A vector $\vec{v} = \langle x, y \rangle$ is a unit vector if and only if $x^2 + y^2 = 1$. Equivalently, using polar parameterization:

$$\vec{v}(t) = \langle \cos t, \sin t \rangle \quad \text{for } 0 \leq t < 2\pi$$



Standard Basis Vectors:

- **2D Basis Vectors:** $\hat{i} = \langle 1, 0 \rangle, \quad \hat{j} = \langle 0, 1 \rangle$
- **3D Basis Vectors:** $\hat{i} = \langle 1, 0, 0 \rangle, \quad \hat{j} = \langle 0, 1, 0 \rangle, \quad \hat{k} = \langle 0, 0, 1 \rangle$

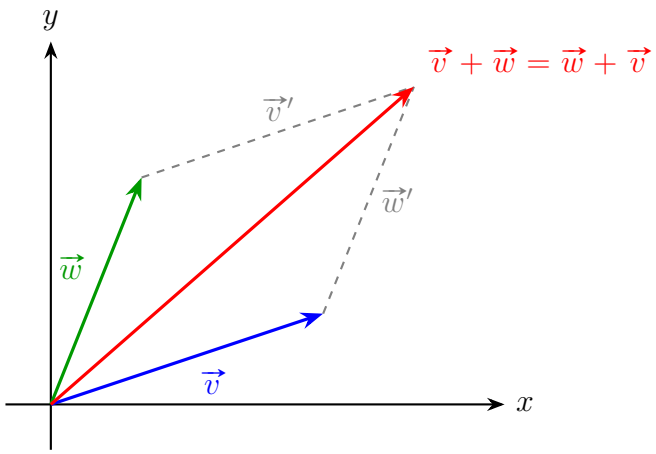
1.7 Vector Operations

Addition and Subtraction:

$$\langle v_1, v_2 \rangle + \langle w_1, w_2 \rangle = \langle v_1 + w_1, v_2 + w_2 \rangle$$
$$\langle v_1, v_2 \rangle - \langle w_1, w_2 \rangle = \langle v_1 - w_1, v_2 - w_2 \rangle \quad \text{(Equivalent to } \vec{v} + (-1)\vec{w}\text{)}$$

Scalar Multiplication: For any scalar $\lambda \in \mathbb{R}$ and vector $\vec{v} = \langle v_1, v_2 \rangle$:

$$\lambda \vec{v} = \langle \lambda v_1, \lambda v_2 \rangle$$



Important Note

Vectors are **not** designed to be divided! (Vector division is undefined).

1.8 Parallel Vectors

Definition 1.7: Parallel Vectors

Two non-zero vectors $\vec{v}$ and $\vec{w}$ ($\vec{v}, \vec{w} \neq \langle 0, 0 \rangle$) are *parallel* (denoted as $\vec{v} \parallel \vec{w}$) if there exists a non-zero scalar $\lambda \in \mathbb{R}$ such that:

$$\vec{w} = \lambda \vec{v}$$

Equivalently, we have a geometric definition.

Definition 1.8: Parallel Vectors (Geometric)

$\vec{v} \parallel \vec{w}$ if and only if both vectors define or lie on the *same line passing through the origin.*

3 Vectors and Directions

Problem 3.1: Vector Directions

Let $\vec{v} = \langle 3, -4, 2 \rangle$. Find a unit vector in the same direction as $\vec{v}$.

Solution 3.1

To find a unit vector in the same direction as $\vec{v}$, we first compute its magnitude:

$$\|\vec{v}\| = \sqrt{3^2 + (-4)^2 + 2^2} = \sqrt{9 + 16 + 4} = \sqrt{29}$$

The unit vector in the same direction as $\vec{v}$ is then:

$$\hat{v} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{1}{\sqrt{29}} \langle 3, -4, 2 \rangle$$

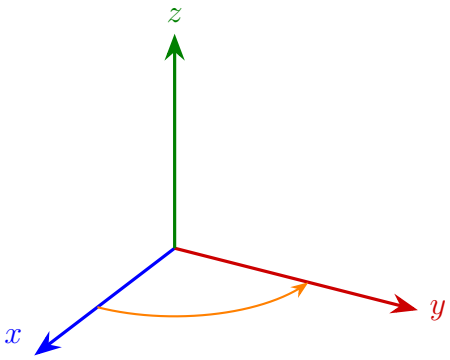
In general, for any non-zero vector $\vec{w}$,

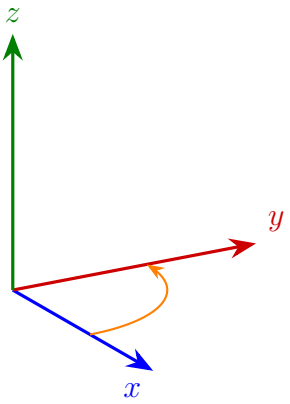
$$\begin{aligned} \|\lambda\vec{w}\| &= \sqrt{(\lambda w_1)^2 + (\lambda w_2)^2 + (\lambda w_3)^2} \\ &= \sqrt{\lambda^2(w_1^2 + w_2^2 + w_3^2)} \\ &= |\lambda|\|\vec{w}\| \end{aligned}$$

Hence, if we want a unit vector in the same direction as $\vec{w}$, we can choose $\lambda = \dfrac{1}{\|\vec{w}\|}$, giving us:

$$\hat{w} = \frac{\vec{w}}{\|\vec{w}\|}$$

Right-Hand Rule The relative position of the x-, y-, and z-axes in $\mathbb{R}^3$ is determined by the *right-hand rule*. If you point your right thumb along the positive x-axis and your index finger along the positive y-axis, then your middle finger (perpendicular to both) will point along the positive z-axis.





Definition 3.1: Parameterizations

A parameterization is a function whose range is a given geometric object.

For example, a parameterization of the unit circle is given by:

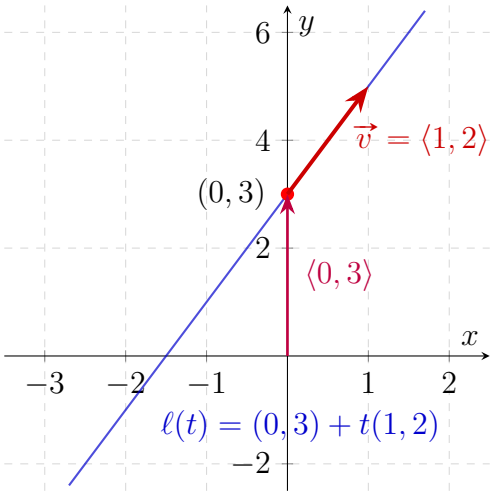
$$f(t) = (\cos t, \sin t)$$

Example 3.1.

To parameterize the line $y = 2x + 3$:

$$\begin{aligned} \ell(t) &= (t, 2t + 3) \\ &= \langle 0, 3 \rangle + \langle t, 2t \rangle \end{aligned}$$

The plot is then given:



To generalize, to parameterize a line in $\mathbb{R}^2$, we can first take a point on the line. Then, we

take some $\vec{v}$ in the direction of the line. Then, we can write the parameterization as:

$$\ell(t) = (p_1 + v_1 t,\ p_2 + v_2 t)$$

In $\mathbb{R}^3$, we can do the same thing.

Example 3.2.

Parameterize the line passing through $P(1, 2, 0)$ and $Q(5, -1, 1)$.

$$\begin{aligned}\overrightarrow{PQ} &= \langle 5-1, -1-2, 1-0\rangle \\ &= \langle 4, -3, 1\rangle \\ \ell(t) &= (1, 2, 0) + t(4, -3, 1) \\ &= (1 + 4t, 2 - 3t, t)\end{aligned}$$

3.1 Dot Product

Definition 3.2: Dot Product

The dot product of two vectors $\vec{v} = \langle v_1, v_2\rangle$ and $\vec{w} = \langle w_1, w_2\rangle$ is defined as:

$$\vec{v} \cdot \vec{w} = v_1 w_1 + v_2 w_2$$

Example 3.3.

$$\vec{v} \cdot \vec{v} = v_1^2 + v_2^2 = \|\vec{v}\|^2$$
$$\vec{i} \cdot \vec{j} = 1 \cdot 0 + 0 \cdot 1 = 0$$

Problem 3.2: Dot Product and Angle

Find pairs of vectors with:

- Dot product > 0
- Dot product $= 0$

- Dot product < 0

Solution 3.2

- $\vec{v} \cdot \vec{v} = \|\vec{v}\|^2 > 0$ for any non-zero vector $\vec{v}$.
- $\vec{i} \cdot \vec{j} = 0$ since they are orthogonal.
- $\langle -1, -1 \rangle \cdot \langle 1, 1 \rangle = -1 - 1 = -2 < 0$.

To conclude, the dot products of any two vectors corresponds to the following situations:

- Dot product $= 0$ if and only if the two vectors are orthogonal (perpendicular).
- Dot product > 0 if the angle between the two vectors is acute (less than 90 degrees).
- Dot product < 0 if the angle between the two vectors is obtuse (greater than 90 degrees).

This is because $\vec{v} \cdot \vec{w} = \|\vec{v}\|\|\vec{w}\| \cos\theta$, where θ is the angle between the vectors. Hence,

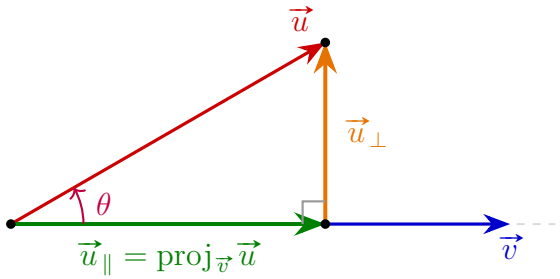
$$\theta = \arccos\left(\frac{\vec{v} \cdot \vec{w}}{\|\vec{v}\|\|\vec{w}\|}\right)$$

Definition 3.3: Orthogonal Projection

Write $\vec{u} = \vec{u}_{\parallel} + \vec{u}_{\perp}$, where $\vec{u}_{\parallel} \parallel \vec{v}$ and $\vec{u}_{\perp} \perp \vec{u}_{\parallel}$. Then, $\vec{u}_{\parallel}$ is called the **orthogonal projection** of $\vec{u}$ onto $\vec{v}$.

Example 3.4.

Below is an illustration of orthogonal projection:



By this, we could see that

$$\vec{u} = \vec{u}_{\parallel} + \vec{u}_{\perp}$$

Then, we solve for $\vec{u}_{\parallel}$ and $\vec{u}_{\perp}$.

Taking the dot product with $\vec{v}$:

$$\begin{aligned}
\vec{u} \cdot \vec{v} &= (\vec{u}_{\parallel} + \vec{u}_{\perp}) \cdot \vec{v} \\
&= \vec{u}_{\parallel} \cdot \vec{v} + \vec{u}_{\perp} \cdot \vec{v} \\
&= \vec{u}_{\parallel} \cdot \vec{v} + 0 \quad (\text{since } \vec{u}_{\perp} \perp \vec{v}) \\
&= \vec{u}_{\parallel} \cdot \vec{v}
\end{aligned}$$

Let $\vec{e}_v = \dfrac{1}{\|\vec{v}\|}\vec{v}$ be the unit vector in the direction of $\vec{v}$, and write $\vec{u}_{\parallel} = \lambda \vec{e}_v$. Then:

$$\begin{aligned}
\vec{u} \cdot \vec{v} &= (\lambda \vec{e}_v) \cdot (\|\vec{v}\| \vec{e}_v) \\
&= \lambda \|\vec{v}\| (\vec{e}_v \cdot \vec{e}_v) \\
&= \lambda \|\vec{v}\| \|\vec{e}_v\|^2 \\
&= \lambda \|\vec{v}\| \quad (\text{since } \|\vec{e}_v\| = 1)
\end{aligned}$$

Solving for λ:

$$\lambda = \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|}$$

Thus, the parallel projection vector $\vec{u}_{\parallel}$ and the orthogonal component $\vec{u}_{\perp}$ are given by:

$$\begin{aligned}
\vec{u}_{\parallel} &= \left(\frac{\vec{u}\cdot\vec{v}}{\|\vec{v}\|}\right)\vec{e}_{v} \\
&= \frac{\vec{u}\cdot\vec{v}}{\|\vec{v}\|^2}\vec{v} \\
&= \frac{\vec{u}\cdot\vec{v}}{\vec{v}\cdot\vec{v}}\vec{v} \\
\vec{u}_{\perp} &= \vec{u} - \vec{u}_{\parallel} \\
&= \vec{u} - \frac{\vec{u}\cdot\vec{v}}{\vec{v}\cdot\vec{v}}\vec{v}
\end{aligned}$$

Furthermore, the magnitude of the parallel projection is:

$$\left\|\vec{u}_{\parallel}\right\| = \|\vec{u}\||\cos\theta|$$

Problem 3.3: Vector Projection

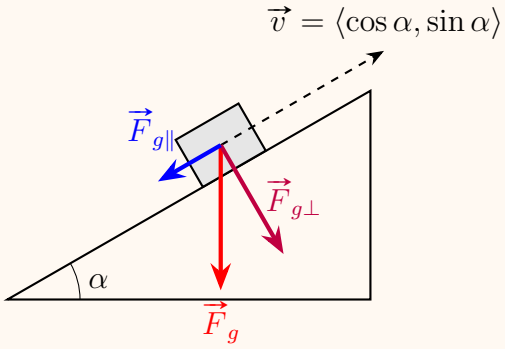
Find the projection of $\vec{u} = \langle 3, 2, 2\rangle$ onto $\vec{v}\,\langle 1, -2, 0\rangle$

Solution 3.3

$$\begin{aligned}
\vec{u}_{\parallel} &= \frac{3\cdot 1 + 2\cdot(-1) + 2\cdot 0}{1^2 + (-1)^2 + 0^2}\langle 1, -1, 0\rangle \\
&= \frac{1}{2}\langle 1, -1, 0\rangle \\
&= \left\langle \frac{1}{2}, -\frac{1}{2}, 0 \right\rangle
\end{aligned}$$

Problem 3.4: Acceleration on an Inclined Plane

Consider a block sliding down an inclined plane with an angle of inclination α.



Solution 3.4

To find the magnitude of acceleration down the incline, we project the gravitational force $\vec{F}_g$ along the incline direction:

$$\begin{aligned}
\text{acceleration} &= \left\| \vec{F}_{g\parallel} \right\| \\
&= \left\| \vec{F}_g \right\| |\cos\theta| \\
&= \left\| \vec{F}_g \right\| \sin\alpha = 9.8 \sin\alpha \quad (\text{m/s}^2)
\end{aligned}$$

Problem 3.5: Vector Projection

Find the projection of $\vec{u} = \langle 3, 2, 2 \rangle$ onto $\vec{v} = \langle 1, -1, 0 \rangle$.

Solution 3.5

Using the vector projection formula:

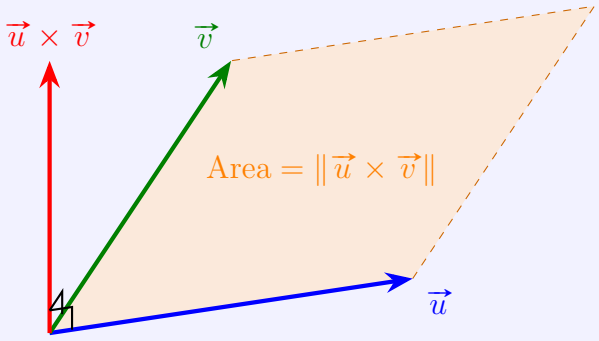
$$\begin{aligned}
\vec{u}_{\parallel} &= \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2} \vec{v} \\
&= \frac{3(1) + 2(-1) + 2(0)}{1^2 + (-1)^2 + 0^2} \langle 1, -1, 0 \rangle \\
&= \frac{3 - 2 + 0}{1 + 1 + 0} \langle 1, -1, 0 \rangle \\
&= \frac{1}{2} \langle 1, -1, 0 \rangle \\
&= \left\langle \frac{1}{2}, -\frac{1}{2}, 0 \right\rangle
\end{aligned}$$

3.2 Determinants and Cross Products

Definition 3.4: Vector Cross Product

Let $\vec{u}, \vec{v} \in \mathbb{R}^3$. The **cross product** $\vec{u} \times \vec{v}$ is a vector in $\mathbb{R}^3$ satisfying the following three geometric properties:

1. **Orthogonality:** $\vec{u} \times \vec{v} \perp \vec{u}$ and $\vec{u} \times \vec{v} \perp \vec{v}$.
2. **Magnitude:** $\|\vec{u} \times \vec{v}\| = \|\vec{u}\|\|\vec{v}\| \sin\theta$, which equals the area of the parallelogram formed by $\vec{u}$ and $\vec{v}$.
3. **Orientation:** The ordered triple $(\vec{u}, \vec{v}, \vec{u} \times \vec{v})$ is right-handed (positively oriented).



From Property (2), we have for the standard basis vectors:

$$\left\|\vec{i} \times \vec{i}\right\| = \left\|\vec{i}\right\|\left\|\vec{i}\right\| \sin 0^\circ = 0 \implies \vec{i} \times \vec{i} = \vec{0}$$

And by the Right-Hand Rule:

$$\vec{i} \times \vec{j} = \vec{k}$$

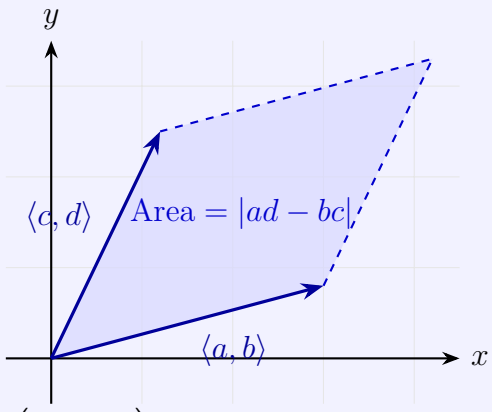
Definition 3.5: Determinants

For a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the determinant is defined as:

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

This is denoted as $\det(A)$ or $|A|$ for a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

For a 2×2 matrix, the determinant represents the area of the parallelogram spanned by $\langle a, c \rangle$ and $\langle b, d \rangle$ (the column vectors) or $\langle a, b \rangle$ and $\langle c, d \rangle$.

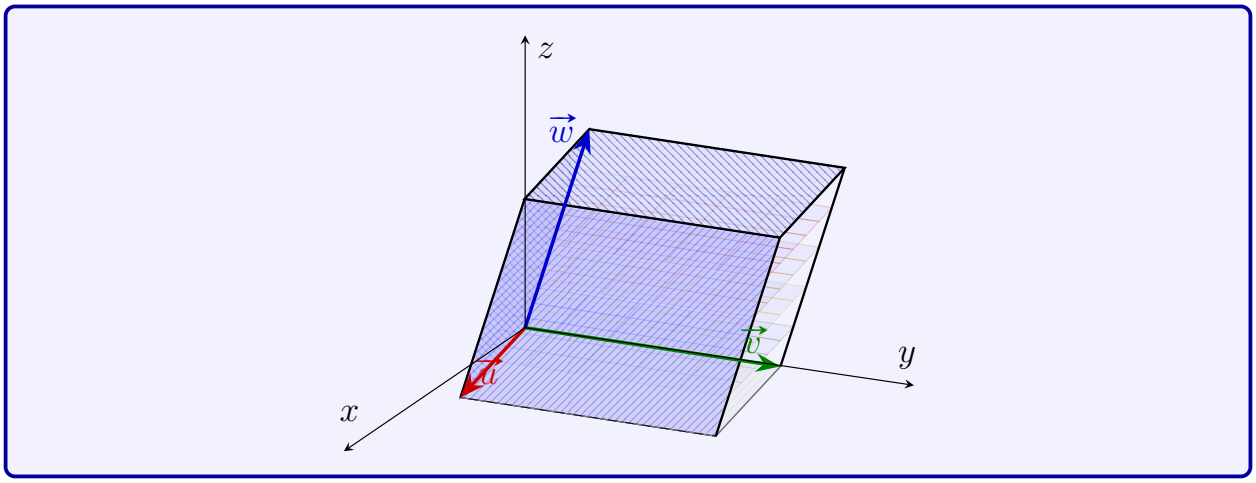


For a 3×3 matrix $B = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$, the determinant is defined as:

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

$$= aei + bfg + cdh - ceg - bdi - afh$$

$\det \begin{pmatrix} \vec{u} \\ \vec{v} \\ \vec{w} \end{pmatrix}$ represents the volume of the parallelepiped spanned by $\vec{u}$, $\vec{v}$, and $\vec{w}$.



The cross product of two vectors $\vec{u} = \langle u_1, u_2, u_3 \rangle$ and $\vec{v} = \langle v_1, v_2, v_3 \rangle$ can be computed using the determinant of a 3×3 matrix:

$$\vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \vec{i} \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} - \vec{j} \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} + \vec{k} \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}$$

Problem 3.6: Cross Product

Let $\vec{v} = \langle 3, 2, -1 \rangle$, $\vec{w} = \langle 4, 1, -3 \rangle$. Find $\vec{v} \times \vec{w}$.

Solution 3.6

$$
\begin{aligned}
\vec{v} \times \vec{w} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 2 & -1 \\ 4 & 1 & -3 \end{vmatrix} \\
&= \vec{i} \begin{vmatrix} 2 & -1 \\ 1 & -3 \end{vmatrix} - \vec{j} \begin{vmatrix} 3 & -1 \\ 4 & -3 \end{vmatrix} + \vec{k} \begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix} \\
&= \vec{i}\,(2(-3) - (-1)(1)) - \vec{j}\,(3(-3) - (-1)(4)) + \vec{k}\,(3(1) - 2(4)) \\
&= \vec{i}\,(-6+1) - \vec{j}\,(-9+4) + \vec{k}\,(3-8) \\
&= \vec{i}\,(-5) - \vec{j}\,(-5) + \vec{k}\,(-5) \\
&= -5\vec{i} + 5\vec{j} - 5\vec{k}
\end{aligned}
$$

3.2.1 Properties of the Cross Product

- $\vec{v} \times \vec{w} \perp \vec{v}$ and $\vec{v} \times \vec{w} \perp \vec{w}$. This implies that $\vec{v} \cdot (\vec{v} \times \vec{w}) = 0$ and $\vec{w} \cdot (\vec{v} \times \vec{w}) = 0$.
- $\{\vec{v}, \vec{w}, \vec{v} \times \vec{w}\}$ forms a right-handed system.
- $\|\vec{v} \times \vec{w}\|$ represents the area of the parallelogram spanned by $\vec{v}$ and $\vec{w}$.

3.2.2 Algebraic Rules of the Cross Product

- Anti-commutativity: $\vec{v} \times \vec{w} = -(\vec{w} \times \vec{v})$
- Parallel vectors: $\vec{v} \times \vec{w} = \vec{0} \iff \vec{v} = \lambda\vec{w}$ or $\vec{w} = \vec{0}$.
- Scalar multiplication: $\lambda(\vec{v} \times \vec{w}) = (\lambda\vec{v}) \times \vec{w} = \vec{v} \times (\lambda\vec{w})$
- Distributivity: $(\vec{u} + \vec{v}) \times \vec{w} = \vec{u} \times \vec{w} + \vec{v} \times \vec{w}$
- Standard basis relations:

$$\vec{i} \times \vec{j} = \vec{k}, \quad \vec{j} \times \vec{k} = \vec{i}, \quad \vec{k} \times \vec{i} = \vec{j}$$

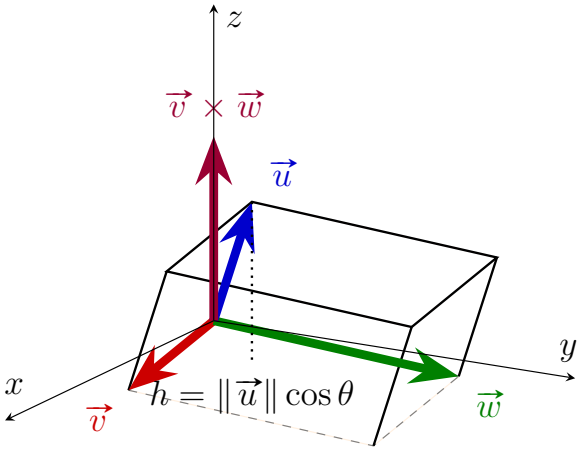
Hence, we may expand the cross product of two vectors $\vec{v} = \langle v_1, v_2, v_3 \rangle$ and $\vec{w} = \langle w_1, w_2, w_3 \rangle$ component-wise:

$$\begin{aligned}
\vec{v} \times \vec{w} &= (v_1 \vec{i} + v_2 \vec{j} + v_3 \vec{k}) \times (w_1 \vec{i} + w_2 \vec{j} + w_3 \vec{k}) \\
&= v_1 w_1 (\vec{i} \times \vec{i}) + v_1 w_2 (\vec{i} \times \vec{j}) + v_1 w_3 (\vec{i} \times \vec{k}) \\
&\quad + v_2 w_1 (\vec{j} \times \vec{i}) + v_2 w_2 (\vec{j} \times \vec{j}) + v_2 w_3 (\vec{j} \times \vec{k}) \\
&\quad + v_3 w_1 (\vec{k} \times \vec{i}) + v_3 w_2 (\vec{k} \times \vec{j}) + v_3 w_3 (\vec{k} \times \vec{k}) \\
&= v_1 w_2 \vec{k} - v_1 w_3 \vec{j} - v_2 w_1 \vec{k} + v_2 w_3 \vec{i} + v_3 w_1 \vec{j} - v_3 w_2 \vec{i} \\
&= (v_2 w_3 - v_3 w_2) \vec{i} + (v_3 w_1 - v_1 w_3) \vec{j} + (v_1 w_2 - v_2 w_1) \vec{k}
\end{aligned}$$

3.3 Scalar Triple Product

Given $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^3$, the quantity $\vec{u} \cdot (\vec{v} \times \vec{w})$ is defined as the **scalar triple product**.

The absolute value $|\vec{u} \cdot (\vec{v} \times \vec{w})|$ gives the geometric volume of the parallelepiped spanned by $\vec{u}$, $\vec{v}$, and $\vec{w}$:



The sign of $\vec{u} \cdot (\vec{v} \times \vec{w})$ indicates the orientation of the vector basis.

$\vec{u} \cdot (\vec{v} \times \vec{w}) > 0$ if the ordered triple $(\vec{u}, \vec{v}, \vec{w})$ forms a **right-handed system** (i.e., $\vec{u}$ points to the same side of the plane spanned by $\vec{v}$ and $\vec{w}$ as $\vec{v} \times \vec{w}$).

4 Planes and Surfaces

4.1 Determining a Plane

A plane $\mathcal{P}$ in $\mathbb{R}^3$ can be uniquely determined by any of the following geometric data:

- A point Q on the plane and a normal vector $\vec{n} \perp \mathcal{P}$.
- Three non-collinear points.
- A general linear equation $ax + by + cz + d = 0$.
- Two intersecting lines (or two non-parallel direction vectors).
- Two parallel distinct lines.
- A line $\mathcal{L}$ and a point $Q \notin \mathcal{L}$.

4.2 Mathematical Formulations of a Plane

Case 1: Plane passing through the origin $O(0, 0, 0)$

A vector $\vec{v} = \langle x, y, z \rangle$ lies in $\mathcal{P} \iff \vec{v} \perp \vec{n}$, where $\vec{n} = \langle a, b, c \rangle$:

$$\vec{v} \in \mathcal{P} \iff \vec{v} \cdot \vec{n} = 0 \iff ax + by + cz = 0$$

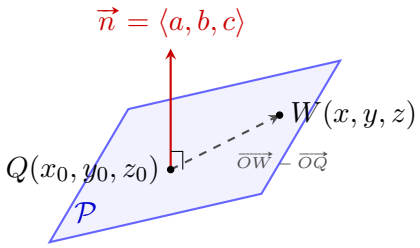
In set notation:

$$\mathcal{P} = \{\langle x, y, z \rangle \mid ax + by + cz = 0\}$$

Case 2: Plane passing through a point $Q(x_0, y_0, z_0)$

Let Q be a known point on $\mathcal{P}$ with position vector $\overrightarrow{OQ} = \langle x_0, y_0, z_0 \rangle$. Any point $W(x, y, z)$ on the plane satisfies:

$$(\overrightarrow{OW} - \overrightarrow{OQ}) \perp \vec{n} \iff (\overrightarrow{OW} - \overrightarrow{OQ}) \cdot \vec{n} = 0$$



Expanding this dot product yields the point-normal form:

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

or equivalently:

$$ax + by + cz = ax_0 + by_0 + cz_0$$

We can always scale the coefficients a, b, c, d in $ax + by + cz = d$ by $\frac{1}{\|\vec{n}\|}$ to obtain a unit normal vector $\hat{n} = \frac{\langle a,b,c \rangle}{\sqrt{a^2+b^2+c^2}}$.

4.3 Properties of Normal Vectors

For any set of vectors $\{\vec{u}, \vec{v}, \vec{w}\}$ where $\vec{v} \nparallel \vec{w}$, the following statements are equivalent:

- $\vec{u} \perp \vec{v}$ and $\vec{u} \perp \vec{w}$.
- $\vec{u} \parallel (\vec{v} \times \vec{w})$.
- $\vec{u} \perp \mathcal{P}$, where $\mathcal{P} = \text{Span}\{\vec{v}, \vec{w}\}$.

Problem 4.1: Finding Normal Vectors

Find a normal vector $\vec{n}$ to the plane $\mathcal{P}$ given by the equation $(x - 1) - (y - 1) = 0$.

Solution 4.1

The equation simplifies to $x - y = 0$. We can extract or compute the normal vector in two ways:

Method 1 (Direct Inspection): From the general form $ax + by + cz = d$, the coefficients of x, y, z directly give the normal vector:

$$\vec{n} = \langle 1, -1, 0 \rangle$$

Method 2 (Cross Product of Parallel Plane Vectors): Two vectors parallel to the plane $\mathcal{P}$ are $\vec{v_1} = \langle 0, 0, 1 \rangle$ and $\vec{v_2} = \langle 1, 1, 0 \rangle$. Taking their cross product gives:

$$\vec{n} = \vec{v_1} \times \vec{v_2} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} = \vec{i}\,(0-1) - \vec{j}\,(0-1) + \vec{k}\,(0-0) = \langle -1, 1, 0 \rangle$$

Note that $\langle -1, 1, 0 \rangle = -\langle 1, -1, 0 \rangle$, which represents the same normal line direction.

Problem 4.2: Finding a Plane From Three Points

Find the equation of the plane containing the points $P(1, 0, 1)$, $Q(0, 2, 0)$, and $R(2, 1, 0)$.

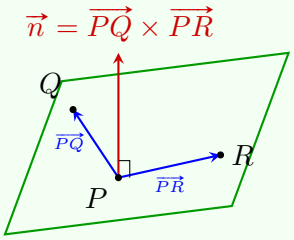
Solution 4.2

Step 1: Construct two displacement vectors in the plane

$$\overrightarrow{PQ} = \langle 0-1, 2-0, 0-1\rangle = \langle -1, 2, -1\rangle$$
$$\overrightarrow{PR} = \langle 2-1, 1-0, 0-1\rangle = \langle 1, 1, -1\rangle$$

Step 2: Compute the normal vector via cross product

$$\vec{n} = \overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -1 & 2 & -1 \\ 1 & 1 & -1 \end{vmatrix} = \langle -1, -2, -3\rangle$$



Step 3: Write the plane equation using point $P(1, 0, 1)$

$$-1(x-1) - 2(y-0) - 3(z-1) = 0$$

Simplifying yields:

$$x + 2y + 3z = 4$$

5 Spheres and Planes

5.1 Determining a Sphere

A sphere in $\mathbb{R}^3$ can be uniquely determined by any of the following sets of data:

- A center point $(a, b, c) \in \mathbb{R}^3$ and a radius $R \in \mathbb{R}^{>0}$.
- Two points representing the endpoints of a diameter.
- Four non-coplanar points.

- A plane tangent to the sphere together with three points lying on the sphere.

5.2 Plane-Sphere Intersections

Example 5.1.

Consider the intersection of the sphere $x^2 + y^2 + z^2 = 4$ with the line $\ell(t) = (t, 0, 1)$ passing through $(1, 0, 1)$ and $(-1, 0, 1)$.

Substituting $\ell(t)$ into the sphere equation yields:

$$\begin{aligned} t^2 + 0^2 + 1^2 &= 4 \\ t^2 + 1 &= 4 \\ t^2 = 3 &\implies t = \pm\sqrt{3} \end{aligned}$$

Thus, the line intersects the sphere at $\ell(\sqrt{3}) = (\sqrt{3}, 0, 1)$ and $\ell(-\sqrt{3}) = (-\sqrt{3}, 0, 1)$.

5.3 Traces of Surfaces

Definition 5.1: Traces

A **trace** of a surface is the intersection curve obtained by intersecting the surface with a parallel coordinate plane of the form $x = k$, $y = k$, or $z = k$.

Example 5.2 (Traces of a Sphere)**.**

Consider the sphere given by:

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = R^2$$

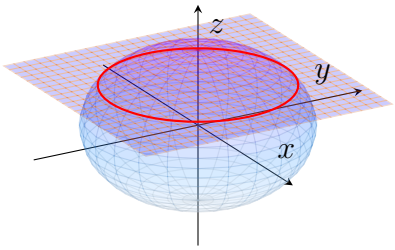
Taking the trace with the plane $z = k$ yields the equation:

$$(x - a)^2 + (y - b)^2 = R^2 - (k - c)^2$$

Depending on the plane distance k, we have three distinct cases:

- **Case 1 ($|k - c| = R$):** The trace is a single point at (a, b, k).

- **Case 2 ($|k-c| < R$):** The trace is a circle of radius $\sqrt{R^2-(k-c)^2}$ centered at (a, b, k).
- **Case 3 ($|k-c| > R$):** The trace is empty (no real intersection).



Example 5.3.

Find the trace of the plane $\mathcal{P} = \{(x, y, z) \mid -2x + 6y - 5z = 4\}$ with the plane $x = -1$:

$$\begin{aligned} -2(-1) + 6y - 5z &= 4 \\ 2 + 6y - 5z &= 4 \\ 6y - 5z = 2 \implies z &= \frac{6}{5}y - \frac{2}{5} \end{aligned}$$

The trace is a straight line in the plane $x = -1$.

5.4 Quadric Surfaces

A general **quadric surface** in $\mathbb{R}^3$ is given by the second-degree equation:

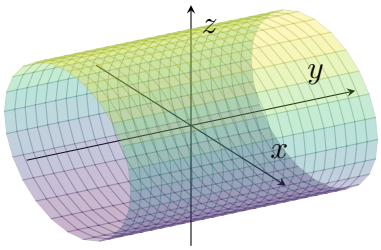
$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fzx + ax + by + cz + d = 0$$

When $D = E = F = 0$, the surface is in **standard position** (axes aligned with coordinate axes).

Example 5.4.

The surface $x^2 + z^2 = 1$ is a circular cylinder aligned along the y-axis with radius 1.

- $z = 0$ trace: $x^2 = 1 \implies x = \pm 1$ (parallel lines along y).
- $x = 0$ trace: $z^2 = 1 \implies z = \pm 1$ (parallel lines along y).

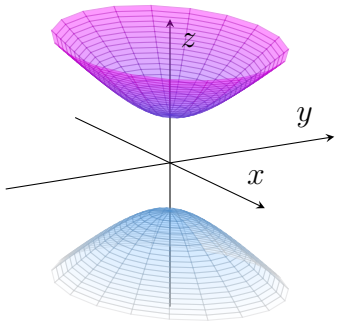


Example 5.5.

Consider the surface given by $x^2 + 4y^2 - z^2 = -1$.

Traces:

- $x = 0$ **trace:** $4y^2 - z^2 = -1 \implies z = \pm\sqrt{4y^2 + 1}$. As $|y| \to \infty$, $z \approx \pm 2y$ (Hyperbola).
- $y = 0$ **trace:** $x^2 - z^2 = -1 \implies z = \pm\sqrt{x^2 + 1}$. As $|x| \to \infty$, $z \approx \pm x$ (Hyperbola).
- $z = k$ **trace:** $x^2 + 4y^2 = k^2 - 1$:
  - $|k| = 1$: Single point at $(0, 0, \pm 1)$.
  - $|k| < 1$: No trace (empty set).
  - $|k| > 1$: Ellipse given by $x^2 + 4y^2 = k^2 - 1$.



Why do cross-sectional ellipses fall on the surface?

For any fixed height $z = k$ with $|k| > 1$, substituting $z = k$ fixes the value $k^2 - 1 > 0$ on the right side of $x^2 + 4y^2 = k^2 - 1$. This traces a locus of points (x, y, k) whose horizontal distance components continuously satisfy the surface equation for that slice.

5.5 Hyperbolic Paraboloid and Its Traces

Definition 5.2: Quadric Surface

A quadric surface is defined by a quadratic equation in three variables x, y, and z:

$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz + ax + by + cz + d = 0$$

, where $A, B, C, D, E, F, a, b, c, d$ are constants, and at least one of A, B, C, D, E, F is non-zero.

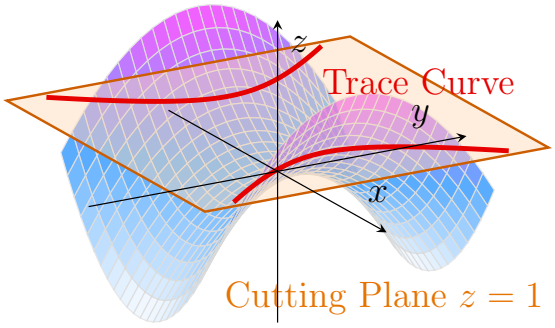
Definition 5.3: Trace

A trace is the intersection of a quadric surface with a plane, usually by taking slices parallel to the coordinate planes.

Note that when our surface can be written by $z = f(x, y)$, the traces form level sets of f.

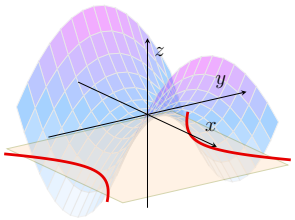
Example 5.6.

Consider the quadratic equation $z = x^2 - y^2$. Now take the traces in $z = -1$, $z = 0$, and $z = 1$. Make a sketch of the traces in $x = -1$, $x = 0$, and $x = 1$. In general, the traces in $x = k$ when k is a constant are hypobolas with equations $z = k^2 - y^2$.

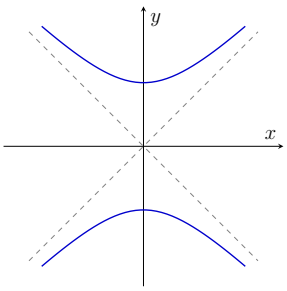


Now take the traces in $z = -1$, $z = 0$, and $z = 1$.

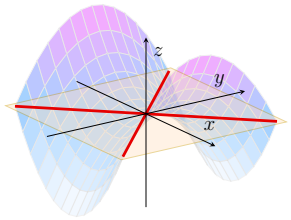
Trace at $z = -1$



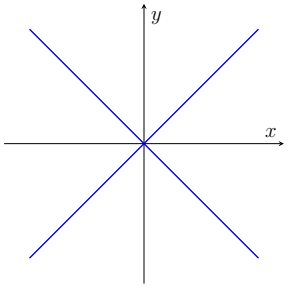
$y^2 - x^2 = 1$



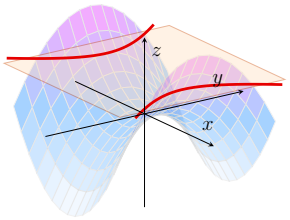
Trace at $z = 0$



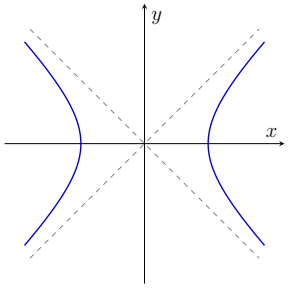
$y = \pm x$



Trace at $z = 1$

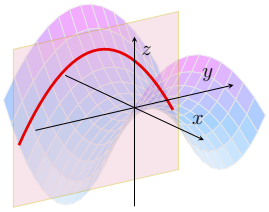


$x^2 - y^2 = 1$

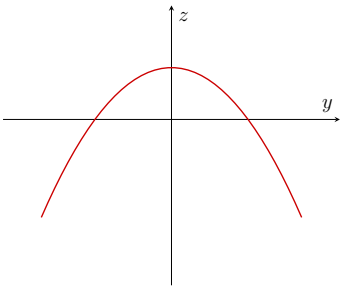


Make a sketch of the traces in $x = -1$, $x = 0$, and $x = 1$. Slicing the surface with vertical planes $x = k$ parallel to the yz-plane yields parabolic trace curves.

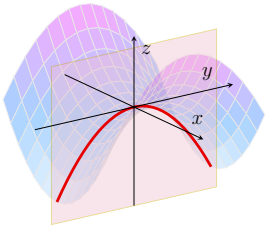
Trace at $x = -1$



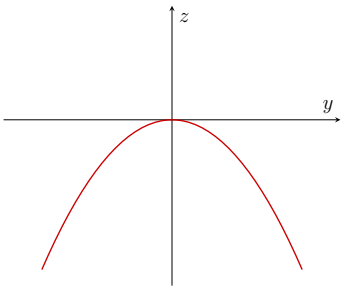
$z = 1 - y^2$



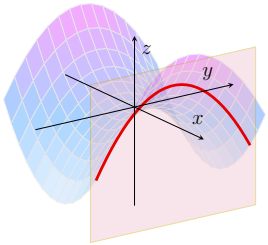
Trace at $x = 0$



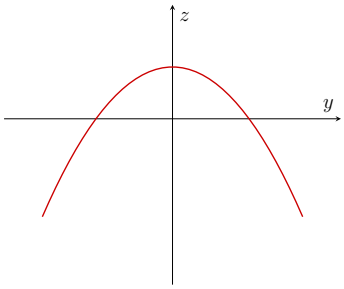
$z = -y^2$



Trace at $x = 1$



$z = 1 - y^2$



Ellipsoids are another family of quadric surfaces. The general equation of an ellipsoid is given by:

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{z}{c}\right)^2 = 1$$

7 Vector-Valued Functions

7.1 Coordinate Systems in 3D Space

Specifying a point or giving directions in space can be accomplished in multiple ways:

- Giving a direction and a distance.
- Using Cartesian coordinates $(d_1, d_2, d_3) = (x, y, z)$.
- Specifying an angle relative to a reference axis along with a magnitude.
- Referencing a relative landmark.
- Using geographic coordinates like latitude and longitude.

7.1.1 Polar Coordinates (2D)

In the xy-plane, a point is specified by its radial distance r from the origin and an angle θ relative to the positive x-axis: (r, θ).

$$r = \sqrt{x^2 + y^2} \quad (r \geq 0)$$

While $\theta = \arctan\left(\frac{y}{x}\right)$ is commonly used, it is undefined when $x = 0$. The foundational algebraic transformation between systems is given by:

$$x = r\cos\theta$$

$$y = r\sin\theta$$

7.1.2 Cylindrical Coordinates

Cylindrical coordinates (r, θ, z) extend polar coordinates into 3D space by preserving the standard Cartesian z-axis:

$$x = r\cos\theta$$
$$y = r\sin\theta$$
$$z = z$$

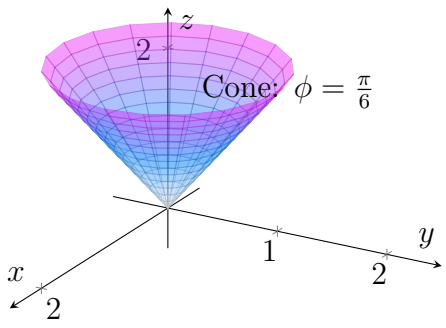
7.1.3 Spherical Coordinates

Spherical coordinates are represented as (ρ, θ, ϕ), where ρ is the distance from the origin, θ is the azimuthal angle in the xy-plane, and ϕ is the polar angle measured down from the positive z-axis.

$$x = \rho\sin\phi\cos\theta$$
$$y = \rho\sin\phi\sin\theta$$
$$z = \rho\cos\phi$$

The standard domain boundaries are:

$$\rho \geq 0$$
$$0 \leq \theta < 2\pi$$
$$0 \leq \phi \leq \pi$$



7.2 Coordinate System Transformations

Problem 7.1: Conversion Between Coordinate Systems

Convert the surface equation $x^2 + \dfrac{y^2}{a^2} = z^2$ into cylindrical and spherical coordinates.

Solution 7.1

Cylindrical Coordinates:

Substituting $x = r\cos\theta$, $y = r\sin\theta$, and $z = z$ into the original equation yields:

$$\begin{aligned}(r\cos\theta)^2 + \frac{(r\sin\theta)^2}{a^2} &= z^2 \\ r^2\cos^2\theta + \frac{r^2\sin^2\theta}{a^2} &= z^2 \\ r^2\left(\cos^2\theta + \frac{\sin^2\theta}{a^2}\right) &= z^2\end{aligned}$$

Spherical Coordinates:

Substituting $x = \rho\sin\phi\cos\theta$, $y = \rho\sin\phi\sin\theta$, and $z = \rho\cos\phi$ into the original equation yields:

$$\begin{aligned}(\rho\sin\phi\cos\theta)^2 + \frac{(\rho\sin\phi\sin\theta)^2}{a^2} &= (\rho\cos\phi)^2 \\ \rho^2\sin^2\phi\cos^2\theta + \frac{\rho^2\sin^2\phi\sin^2\theta}{a^2} &= \rho^2\cos^2\phi \\ \rho^2\sin^2\phi\left(\cos^2\theta + \frac{\sin^2\theta}{a^2}\right) &= \rho^2\cos^2\phi\end{aligned}$$

Assuming $\rho \neq 0$, we divide both sides by ρ^2:

$$\sin^2\phi\left(\cos^2\theta + \frac{\sin^2\theta}{a^2}\right) = \cos^2\phi$$
$$\cos^2\theta + \frac{\sin^2\theta}{a^2} = \frac{\cos^2\phi}{\sin^2\phi}$$
$$\cos^2\theta + \frac{\sin^2\theta}{a^2} = \cot^2\phi$$

7.3 Vector-Valued Functions

Definition 7.1: Vector-Valued Functions

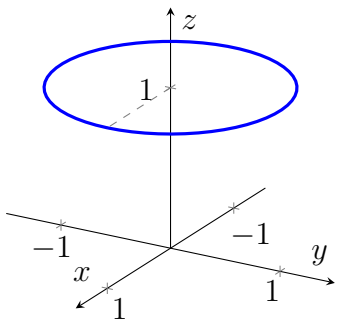
A vector-valued function maps one or more real parameters to a space vector:

$$\vec{r}(t) = \langle x(t), y(t), z(t)\rangle\,, \quad \vec{r} : I \to \mathbb{R}^3$$

where $I \subseteq \mathbb{R}$ is an interval. This traces out a spatial curve parametrized by t.

Example 7.1.

Consider the vector function $\vec{r}(t) = \langle \cos t, \sin t, 1\rangle$ for $0 \leq t < 2\pi$. This describes a unit circle in 3D space elevated to the horizontal plane $z = 1$.



Problem 7.2: Finding Collisions and Intersections

Let $\vec{r}_1(t) = \left\langle t^2 + \frac{t}{2}, t - t^3 \right\rangle$ and $\vec{r}_2(t) = \langle t, t^2\rangle$. Find all collision points and structural intersections between the curves.

Solution 7.2

1. Collision Points (Same location at same time t):

Setting $\vec{r}_1(t) = \vec{r}_2(t)$ gives:

$$\left\langle t^2 + \frac{t}{2}, t - t^3 \right\rangle = \langle t, t^2 \rangle$$

Equating the individual components gives the system:

$$t^2 + \frac{t}{2} = t$$
$$t - t^3 = t^2$$

Solving the first equation for t:

$$t^2 + \frac{t}{2} - t = 0$$
$$t^2 - \frac{t}{2} = 0$$
$$t\left(t - \frac{1}{2}\right) = 0 \implies t = 0 \quad \text{or} \quad t = \frac{1}{2}$$

Now, we verify both parameters in the second component equation $t - t^3 = t^2$:

$$\text{For } t = 0: \quad 0 - (0)^3 = (0)^2 \implies 0 = 0 \quad \text{(Valid)}$$
$$\text{For } t = \frac{1}{2}: \quad \frac{1}{2} - \left(\frac{1}{2}\right)^3 = \frac{1}{2} - \frac{1}{8} = \frac{3}{8}$$
$$\left(\frac{1}{2}\right)^2 = \frac{1}{4} = \frac{2}{8} \neq \frac{3}{8} \quad \text{(Invalid)}$$

Therefore, $t = 0$ is the only valid collision time. Evaluating $\vec{r}_1(0)$:

$$\vec{r}_1(0) = \left\langle 0^2 + \frac{0}{2}, 0 - 0^3 \right\rangle = \langle 0, 0 \rangle$$

The collision point is $(0, 0)$.

2. Intersection Points (Same location at potentially different times t and

s**):**

Setting $\vec{r}_1(t) = \vec{r}_2(s)$ gives:

$$\left\langle t^2 + \frac{t}{2}, t - t^3 \right\rangle = \left\langle s, s^2 \right\rangle$$

Equating components yields:

$$s = t^2 + \frac{t}{2} \tag{1}$$

$$t - t^3 = s^2 \tag{2}$$

Substituting equation (1) into equation (2):

$$\begin{aligned} t - t^3 &= \left(t^2 + \frac{t}{2}\right)^2 \\ t - t^3 &= t^4 + t^3 + \frac{t^2}{4} \\ 0 &= t^4 + 2t^3 + \frac{t^2}{4} - t \\ 0 &= t\left(t^3 + 2t^2 + \frac{t}{4} - 1\right) \end{aligned}$$

Setting each factor to zero:

$$\begin{aligned} t = 0 &\implies s = 0^2 + \frac{0}{2} = 0 \\ &\implies \text{Intersection point } (0, 0) \\ t^3 + 2t^2 + \frac{t}{4} - 1 = 0 &\implies t \approx 0.576 \end{aligned}$$

Solving for s when $t \approx 0.576$:

$$\begin{aligned} s &= (0.576)^2 + \frac{0.576}{2} \\ s &\approx 0.332 + 0.288 \approx 0.620 \end{aligned}$$

Evaluating the spatial coordinates using $\vec{r}_2(s)$:

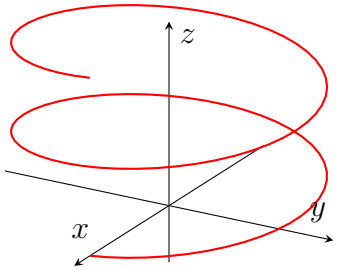
$$x = s \approx 0.620$$

$$y = s^2 \approx (0.620)^2 \approx 0.384$$

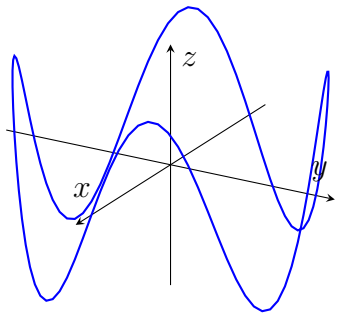
Hence, the two spatial intersection points are $(0, 0)$ and $(0.620, 0.384)$.

7.4 Gallery of Parametric Space Curves

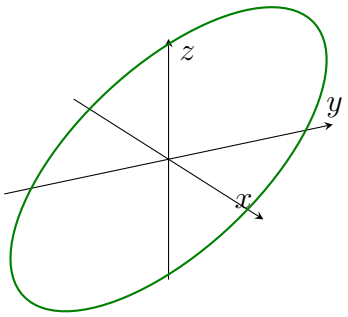
Below are standard parametric 3D space curves used in multivariable trajectory analysis:



$\vec{r}(t) = \langle \cos t, \sin t, t \rangle$



$\vec{r}(s) = \langle \cos s, \sin s, \sin(4s) \rangle$



$\vec{r}(t) = \langle \cos t, \sin t, 4 \sin t \rangle$

7.5 Parameterization of Curves and Surfaces

Problem 7.3: Parameterization of a Curve on a Surface

Find a parameterization for the curve given by the intersection of the surfaces $y = \sqrt{x^2 + z^2}$ and $x = z^2$.

Solution 7.3

Notice that for any choice of $z \in \mathbb{R}$, the value of x is uniquely determined by $x = z^2$.

We can set $z = t$ as our parameter. Then,

$$x = t^2$$

$$z = t$$

$$y = \sqrt{(t^2)^2 + t^2} = \sqrt{t^4 + t^2} = |t|\sqrt{t^2 + 1}$$

Thus, a vector parameterization of the curve is:

$$\vec{r}(t) = \left\langle t^2, |t|\sqrt{t^2+1}, t \right\rangle, \quad t \in \mathbb{R}$$

7.6 Calculus of Vector-Valued Functions

Calculus operations on vector-valued functions are defined component-wise.

For a vector-valued function $\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$, limits, derivatives, and integrals are computed as follows:

$$\lim_{t \to t_0} \vec{r}(t) = \left\langle \lim_{t \to t_0} x(t), \lim_{t \to t_0} y(t), \lim_{t \to t_0} z(t) \right\rangle$$

$$\vec{r}'(t) = \langle x'(t), y'(t), z'(t) \rangle$$

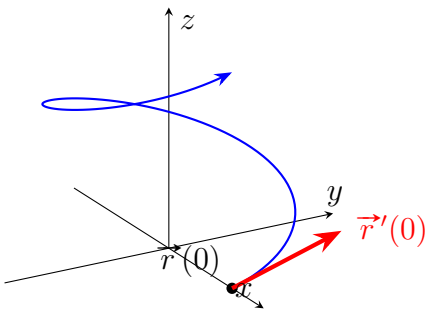
$$\int_a^b \vec{r}(t)\mathrm{d}t = \left\langle \int_a^b x(t)\mathrm{d}t, \int_a^b y(t)\mathrm{d}t, \int_a^b z(t)\mathrm{d}t \right\rangle$$

Example 7.2 (Tangent Vectors and Parameterizing Tangent Lines)**.**
Given the circular helix $\vec{r}(t) = \langle \cos t, \sin t, t \rangle$.

Differentiating component-wise gives:

$$\vec{r}'(t) = \langle -\sin t, \cos t, 1 \rangle$$
$$\vec{r}'(0) = \langle 0, 1, 1 \rangle$$

The derivative $\vec{r}'(0)$ yields a vector representing the tangent direction of the curve at $t = 0$.



To parameterize the tangent line $\ell(t)$ to the helix at $t = 0$:

$$\vec{r}(0) = \langle 1, 0, 0 \rangle$$

Thus,

$$\begin{aligned} \ell(t) &= \vec{r}(0) + t\,\vec{r}'(0) \\ &= \langle 1, 0, 0 \rangle + t\,\langle 0, 1, 1 \rangle = \langle 1, t, t \rangle \end{aligned}$$

Problem 7.4: Derivative and Singular Points of a Cycloid

Find the derivative of the vector-valued function $\vec{r}(t) = \langle t - \sin t, 1 - \cos t \rangle$ and evaluate it at $t = \frac{\pi}{2}, \pi, 2\pi$.

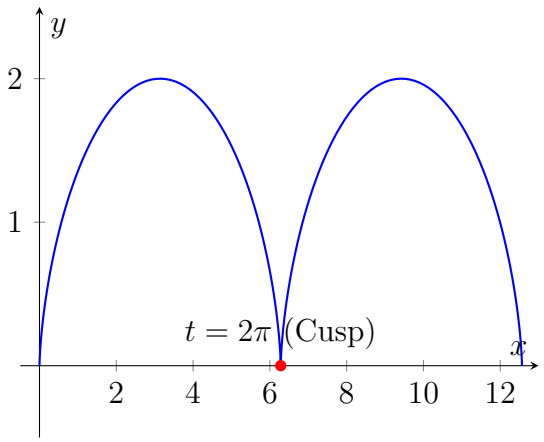
Solution 7.4

Differentiating each component with respect to t:

$$\begin{aligned} \vec{r}'(t) &= \left\langle \frac{\mathrm{d}}{\mathrm{d}t}(t - \sin t), \frac{\mathrm{d}}{\mathrm{d}t}(1 - \cos t) \right\rangle \\ &= \langle 1 - \cos t, \sin t \rangle \end{aligned}$$

Evaluating at specific parameter values:

$$\vec{r}'\left(\frac{\pi}{2}\right) = \left\langle 1 - \cos\frac{\pi}{2}, \sin\frac{\pi}{2} \right\rangle = \langle 1, 1 \rangle$$
$$\vec{r}'(\pi) = \langle 1 - \cos\pi, \sin\pi \rangle = \langle 2, 0 \rangle$$
$$\vec{r}'(2\pi) = \langle 1 - \cos 2\pi, \sin 2\pi \rangle = \langle 0, 0 \rangle$$



Notice from the graph of the cycloid that at $t = 2\pi$, the curve forms a sharp sharp corner (cusp). At this point, $\vec{r}'(2\pi) = \langle 0, 0 \rangle$. Although the vector function $\vec{r}(t)$ is differentiable everywhere, the tangent line direction is undefined at the cusp because the derivative vector vanishes.

7.7 Properties of Vector Derivatives & Integration

Problem 7.5: Orthogonality Property for Constant-Length Vector Functions

If $\|\vec{r}(t)\| = c$ for some constant c, what does this imply about the relationship between $\vec{r}(t)$ and its derivative $\vec{r}'(t)$?

Solution 7.5

Since $\|\vec{r}(t)\| = c$, taking the square of both sides yields:

$$\|\vec{r}(t)\|^2 = c^2 \implies \vec{r}(t) \cdot \vec{r}(t) = c^2$$

Differentiating both sides with respect to t using the product rule for dot products:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\vec{r}(t)\cdot\vec{r}(t)\right)=\frac{\mathrm{d}}{\mathrm{d}t}(c^2)$$
$$\vec{r}\,'(t)\cdot\vec{r}(t)+\vec{r}(t)\cdot\vec{r}\,'(t)=0$$
$$2\vec{r}(t)\cdot\vec{r}\,'(t)=0 \implies \vec{r}(t)\cdot\vec{r}\,'(t)=0$$

Therefore, $\vec{r}(t)\perp\vec{r}\,'(t)$ for all t. (The position vector is always orthogonal to its velocity vector).

Product Rule Verification for Dot Products

Given two vector-valued functions $\vec{r_1}(t),\vec{r_2}(t)\in\mathbb{R}^2$:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left[\vec{r_1}(t)\cdot\vec{r_2}(t)\right]&=\frac{\mathrm{d}}{\mathrm{d}t}\left[x_1(t)x_2(t)+y_1(t)y_2(t)\right]\\&=x_1'(t)x_2(t)+x_1(t)x_2'(t)+y_1'(t)y_2(t)+y_1(t)y_2'(t)\\&=\vec{r_1}\,'(t)\cdot\vec{r_2}(t)+\vec{r_1}(t)\cdot\vec{r_2}\,'(t)\end{aligned}$$

Problem 7.6: Initial Value Problem for Motion

A particle moves with velocity $\vec{v}(t)=\left\langle 3t^2-2,\cos(\pi t),\sqrt{t}\right\rangle$. Find its position vector $\vec{r}(t)$, given the initial condition $\vec{r}(0)=\langle 2,1,1\rangle$.

Solution 7.6

Integrating $\vec{v}(t)$ component-wise:

$$\begin{aligned}\vec{r}(t)&=\int\vec{v}(t)\,\mathrm{d}t\\&=\left\langle\int(3t^2-2)\,\mathrm{d}t,\int\cos(\pi t)\,\mathrm{d}t,\int t^{1/2}\,\mathrm{d}t\right\rangle\\&=\left\langle t^3-2t+C_1,\frac{1}{\pi}\sin(\pi t)+C_2,\frac{2}{3}t^{3/2}+C_3\right\rangle\end{aligned}$$

Applying the initial condition $\vec{r}(0) = \langle 2, 1, 1\rangle$:

$$\vec{r}(0) = \langle C_1, C_2, C_3\rangle = \langle 2, 1, 1\rangle \implies C_1 = 2,\, C_2 = 1,\, C_3 = 1$$

Therefore,

$$\vec{r}(t) = \left\langle t^3 - 2t + 2, \frac{1}{\pi}\sin(\pi t) + 1, \frac{2}{3}t^{3/2} + 1 \right\rangle$$

Note on Circular Motion: If given acceleration $\vec{r}''(t) = \langle -\cos t, -\sin t, 0\rangle$, to uniquely guarantee that $\vec{r}(t)$ describes a circle centered at the origin, appropriate initial conditions for both position $\vec{r}(0)$ and velocity $\vec{r}'(0)$ must be specified.

7.8 Arc Length of Curves

The arc length s of a smooth curve $\vec{r}(t) = \langle x(t), y(t)\rangle$ on $t \in [a, b]$ is derived as the limit of polygonal approximations:

$$s = \lim_{\Delta t \to 0} \sum \sqrt{\left(\frac{\Delta x}{\Delta t}\right)^2 + \left(\frac{\Delta y}{\Delta t}\right)^2}\,\Delta t = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2}\,\mathrm{d}t = \int_a^b \|\vec{r}'(t)\|\mathrm{d}t$$

Example 7.3 (Arc Length of a Helix)**.**
Find the arc length of $\vec{r}(t) = \langle \cos t, \sin t, t\rangle$ for $t \in [0, 2\pi]$.

First, evaluate the speed $\|\vec{r}'(t)\|$:

$$\vec{r}'(t) = \langle -\sin t, \cos t, 1\rangle \implies \|\vec{r}'(t)\| = \sqrt{(-\sin t)^2 + \cos^2 t + 1^2} = \sqrt{2}$$

Thus, the arc length is:

$$s = \int_0^{2\pi} \sqrt{2}\,\mathrm{d}t = 2\pi\sqrt{2}$$

Example 7.4 (Arc Length of an Ellipse and Elliptic Integrals)**.**
Consider the ellipse $x^2 + \frac{y^2}{4} = 1$.

Method 1 (Cartesian Parameterization - Upper Half): Setting $x = t$ and $y =$

$2\sqrt{1-t^2}$ for $t \in [-1, 1]$ yields:

$$\vec{r}(t) = \left\langle t, 2\sqrt{1-t^2} \right\rangle \implies \vec{r}\,'(t) = \left\langle 1, -\frac{2t}{\sqrt{1-t^2}} \right\rangle$$

$$\|\vec{r}\,'(t)\| = \sqrt{1 + \frac{4t^2}{1-t^2}} = \sqrt{\frac{1+3t^2}{1-t^2}}$$

The arc length of the upper half is given by

$$\int_{-1}^{1} \sqrt{\frac{1+3t^2}{1-t^2}}\, \mathrm{d}t.$$

Method 2 (Trigonometric Parameterization - Full Ellipse): Setting $x = \cos t$ and $y = 2\sin t$ for $t \in [0, 2\pi]$:

$$\vec{r}\,'(t) = \langle -\sin t, 2\cos t \rangle \implies \|\vec{r}\,'(t)\| = \sqrt{\sin^2 t + 4\cos^2 t} = \sqrt{4 - 3\sin^2 t}$$

The total perimeter of the ellipse is:

$$s = \int_0^{2\pi} \sqrt{4 - 3\sin^2 t}\, \mathrm{d}t$$

Definition 7.2: Elliptic Integral

An elliptic integral is a type of integral that cannot be expressed in terms of elementary functions.

Elliptic Integrals: It turns out these integrals cannot be evaluated in terms of elementary functions. They belong to a class of functions known as **elliptic integrals**, which require numerical methods to evaluate.

8 Scalar-Valued Functions of Several Variables

Definition 8.1: Scalar Function

A scalar-valued function of several variables maps a vector of multiple inputs to a single real output $f : D \subseteq \mathbb{R}^n \to \mathbb{R}$.

Examples:

- Temperature distribution: $T(\text{longitude}, \text{latitude}, \text{time}) : \mathbb{R}^3 \to \mathbb{R}$.
- Precipitation depth: $P(\text{longitude}, \text{latitude}) : \mathbb{R}^2 \to \mathbb{R}$.
- Domain and Range example: For $f(x, y) = y\sqrt{x-1}$,

$$\text{Domain: } \{(x, y) \in \mathbb{R}^2 : x \geq 1\}, \quad \text{Range: } \mathbb{R}$$

8.1 Visualizing Multivariable Functions: Traces and Level Sets

The graph of a function $f(x, y)$ is defined as the surface $\{(x, y, z) \in \mathbb{R}^3 : z = f(x, y)\}$.

Example 8.1 (Analyzing Traces of $f(x, y) = y \sin x$)**.**

To visualize $f(x, y) = y \sin x$, we examine slices along constant coordinates (traces):

- **y-traces ($y = c$):**

  For $y = 0 \implies z = 0$;

  for $y = 1 \implies z = \sin x$;

  for $y = -1 \implies z = -\sin x$. (Sinusoidal curves scaled by y).

- **x-traces ($x = c$):**

  For $x = 0 \implies z = 0$;

  for $x = \frac{\pi}{2} \implies z = y$;

  for $x = \pi \implies z = 0$. (Lines whose slopes depend on $\sin x$).

8.2 Level Sets and Contour Maps

Definition 8.2: Level Set

For a function $f : D \to \mathbb{R}$ ($D \subseteq \mathbb{R}^2$), the **level set** (or level curve) at height c is defined as

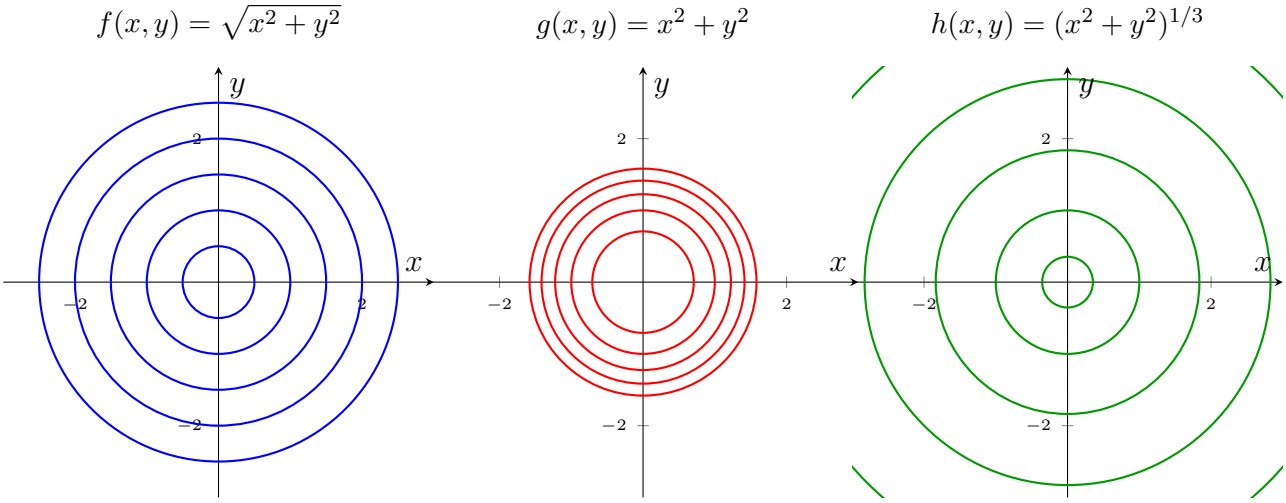
$$\{(x, y) \in \mathbb{R}^2 : f(x, y) = c\}.$$

A collection of such level curves forms a **contour map**.

Example 8.2 (Comparing Level Curves)**.**

Consider three functions and their level sets $f(x, y) = c$:

1. $f(x, y) = \sqrt{x^2 + y^2}$: Level sets are equally spaced concentric circles $x^2 + y^2 = c^2$ (a cone).
2. $g(x, y) = x^2 + y^2$: Level sets are circles $x^2 + y^2 = c$ spaced closer together as c increases (a paraboloid).
3. $h(x, y) = (x^2 + y^2)^{1/3}$: Level sets are circles $x^2 + y^2 = c^3$ spaced further apart as c increases.



8.3 Visualizing Higher Dimensions via Slices

- A sphere in $\mathbb{R}^3$ defined by $x^2 + y^2 + z^2 = \rho^2$ can be sliced along $z = c$ to produce 2D circular cross-sections in $\mathbb{R}^2$.

- A 3-sphere in $\mathbb{R}^4$ defined by $x^2 + y^2 + z^2 + w^2 = R^2$ can be visualized through its 3D cross-sections ($w = c$), which appear as concentric 2-spheres in $\mathbb{R}^3$ whose radii expand from 0 to R and then contract back to 0.

3D Sphere Slices ($z = c$) 4D Hypersphere Slices ($w = c$ in $\mathbb{R}^3$)



8.4 Limits of Functions of Several Variables

For single-variable functions, we say that the limit of a function $f(x)$ exists such that

$$\lim_{x \to a} f(x) = L \iff \forall \epsilon > 0, \exists \delta > 0 \text{ such that } 0 < |x - a| < \delta \implies |f(x) - L| < \epsilon$$

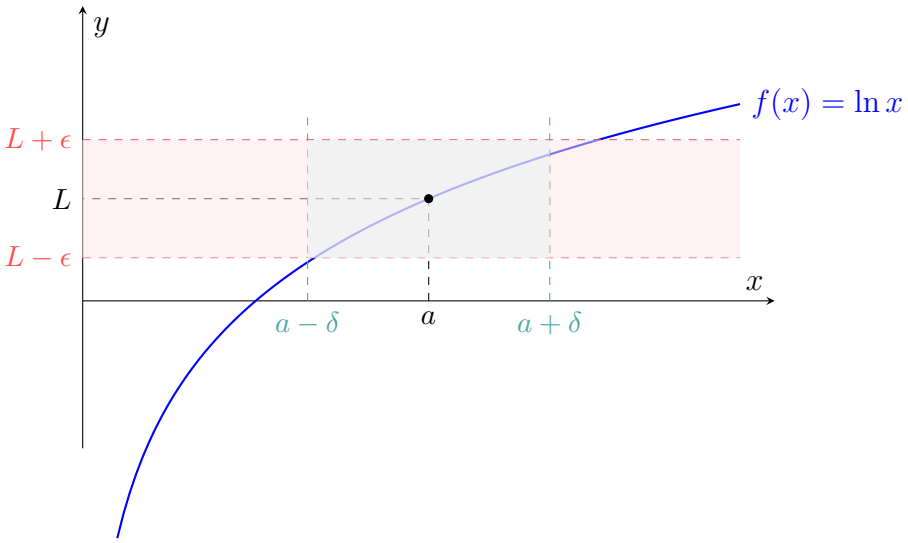
As shown below, we can visualize the ϵ-δ definition of a limit for a single-variable function $f(x) = \ln x$ at $x = a$.

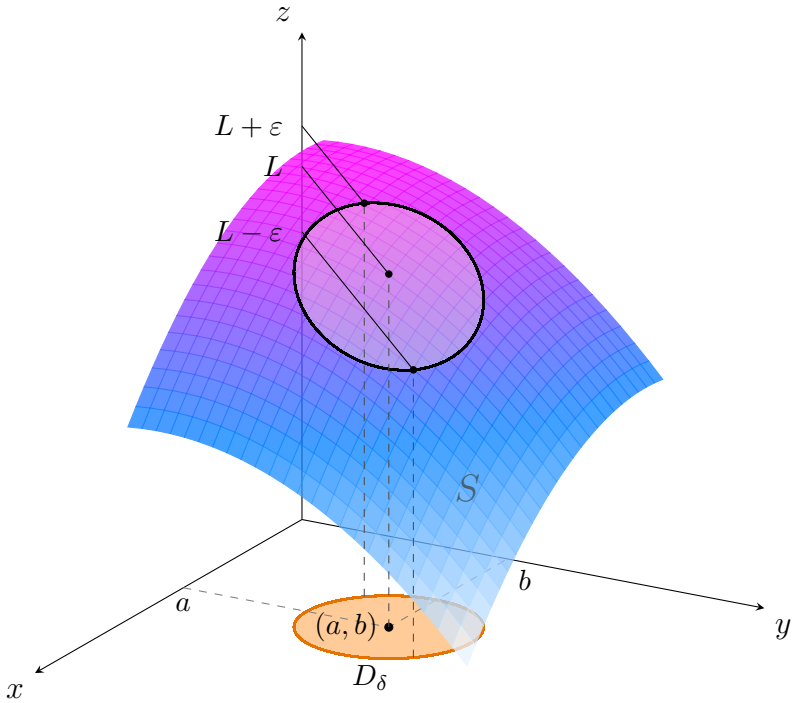
Now consider the case of functions of two variables. We say that the limit of a function $f(x, y)$ exists such that

$$\lim_{(x,y) \to (a,b)} f(x, y) = L \iff \forall \epsilon > 0, \exists \delta > 0 \text{ such that } 0 < \sqrt{(x - a)^2 + (y - b)^2} < \delta$$
$$\implies |f(x, y) - L| < \epsilon$$

Similarly, we can visualize the ϵ-δ definition of a limit for a two-variable function $f(x, y)$.

ϵ-δ Definition of Limit for $f(x) = \ln x$ at $x = a$.





Problem 8.1: Limit of a Function of Two Variables

Find, if it exists, $\lim_{(x,y)\to(0,0)} \dfrac{xy}{x^2+y^2}$.

Solution 8.1

First, we consider approaching $(0,0)$ along the x-axis, i.e., $y=0$:

$$\lim_{x\to 0} \frac{x(0)}{x^2+0^2} = 0$$

Next, we consider approaching $(0,0)$ along the y-axis, i.e., $x=0$:

$$\lim_{y\to 0} \frac{0(y)}{0^2+y^2} = 0$$

Next, we consider approaching $(0,0)$ along the line $y=x$:

$$\lim_{x\to 0} \frac{x(x)}{x^2+x^2} = \lim_{x\to 0} \frac{x^2}{2x^2} = \lim_{x\to 0} \frac{1}{2} = \frac{1}{2}$$

Since approaching along different paths yields different limiting values, the limit does not exist.

Systematic Path Approaches Checking paths individually can be cumbersome. Instead, we can systematically analyze general path families:

1. Linear Paths ($y=mx$):

$$\lim_{x\to 0} f(x,mx) = \lim_{x\to 0} \frac{x(mx)}{x^2+(mx)^2} = \lim_{x\to 0} \frac{mx^2}{x^2(1+m^2)} = \frac{m}{1+m^2}$$

The limit depends explicitly on the slope m. Hence, the limit does not exist.

2. Polar Coordinates ($x=r\cos\theta, y=r\sin\theta$):

$$\lim_{r\to 0} f(r\cos\theta, r\sin\theta) = \lim_{r\to 0} \frac{r^2\cos\theta\sin\theta}{r^2} = \cos\theta\sin\theta$$

The limit depends on the angle θ, confirming the limit does not exist.

Definition 8.3: Squeeze Theorem for Two Variables

If $g(x,y) \leq f(x,y) \leq h(x,y)$ for all (x,y) in a disk centered at (a,b) (except possibly at (a,b) itself), and

$$\lim_{(x,y)\to(a,b)} g(x,y) = \lim_{(x,y)\to(a,b)} h(x,y) = L$$

then $\lim_{(x,y)\to(a,b)} f(x,y) = L$.

Definition 8.4: Continuity of Functions of Several Variables

A function $f(x,y)$ is *continuous* at (a,b) if:

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$$

Furthermore, $f(x,y)$ is continuous on a domain D if it is continuous at every point $(a,b) \in D$.

Example 8.3 (Continuity of a Composite Function)**.**
Consider $f(x,y) = \arctan\left(\frac{y}{x}\right)$.

The inner function $g(x,y) = \frac{y}{x}$ is continuous everywhere except on the line $x = 0$. Since $h(t) = \arctan(t)$ is continuous on $\mathbb{R}$, the composite function $f(x,y) = h(g(x,y))$ is continuous at all points $(x,y) \in \mathbb{R}^2$ where $x \neq 0$.

9 Partial Derivatives

9.1 Definition and Properties of Partial Derivatives

Definition 9.1: Partial Derivative

In this course, we will focus on functions of two variables, i.e., $f(x, y)$. For a binary function $f(x, y)$, the partial derivative with respect to x is given by:

$$\frac{\partial f}{\partial x} = \lim_{h \to 0} \frac{f(x + h, y) - f(x, y)}{h}$$

and the partial derivative with respect to y is given by:

$$\frac{\partial f}{\partial y} = \lim_{h \to 0} \frac{f(x, y + h) - f(x, y)}{h}$$

provided these limits exist.

In general, let $f(x_1, x_2, \ldots, x_n)$ be a function of n variables. The **partial derivative** of f with respect to the variable x_i is defined as:

$$\frac{\partial f}{\partial x_i} = \lim_{h \to 0} \frac{f(x_1, x_2, \ldots, x_i + h, \ldots, x_n) - f(x_1, x_2, \ldots, x_i, \ldots, x_n)}{h}$$

provided this limit exists.

Some key points to note about partial derivatives:

- $f_x(a, b)$ is the instantaneous rate of change of f at the point (a, b) in the x-direction.
- $f_x(a, b)$ is the slope of the tangent line to $f(a, b)$ at the point $x = a$.

Computing partial derivatives is similar to computing derivatives in single-variable calculus. The only difference is that we treat all other variables as constants when differentiating with respect to one variable.

Example 9.1 (Partial Derivatives of a Binary Function)**.**

Let $f(x, y) = e^{2x} \cos(y)$.

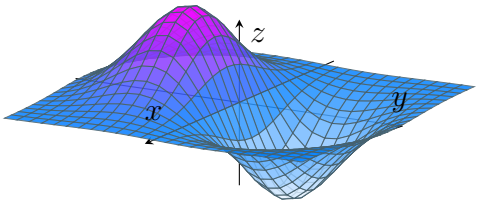
We can now compute the partial derivatives of f with respect to x and y:

$$f_x(x,y) = \frac{\partial}{\partial x}\left(e^{2x}\cos(y)\right) = 2e^{2x}\cos(y)$$
$$f_y(x,y) = \frac{\partial}{\partial y}\left(e^{2x}\cos(y)\right) = -e^{2x}\sin(y)$$

For better comprehension, we can visualize the partial derivatives of a function of two variables as the slopes of tangent lines to the surface defined by the function in the x and y directions.

Example 9.2 (Visualizing Partial Derivatives)**.**

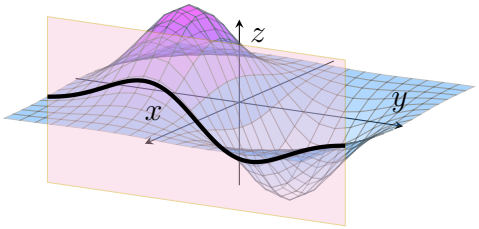
Consider the surface defined by $f(x,y) = -\frac{10y}{e^{x^2+y^2}}$ and its slices along fixed coordinate planes.

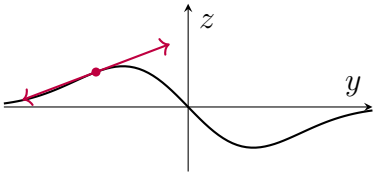


(Original surface $z = f(x,y)$)

$x = 1$ **Trace:**

$$z = f(1,y) = -\frac{10y}{e^{1+y^2}}$$

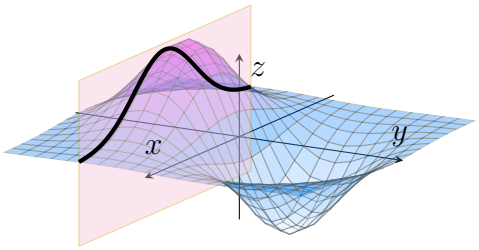




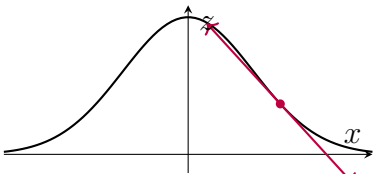
At $(1,-1)$,

$$f_y(1,-1) = \frac{10}{e^2} > 0.$$

$y = -1$ **Trace:**



$$z = f(x, -1) = \frac{10}{e^{x^2+1}}$$

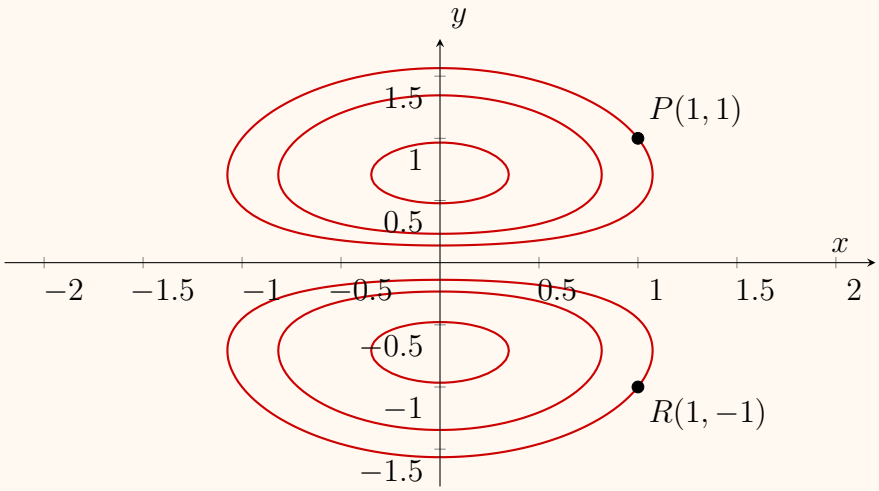


At $(1, -1)$,

$$f_x(1, -1) = -\frac{20}{e^2} < 0.$$

Problem 9.1: Visualizing Partial Derivatives on a Contour Map

Give some reasonable labelings to the contour map based on the image of the graph above.



Determine whether the following values are **positive**, **negative**, or **0**:

1. $f_x(O)$ and $f_y(O)$ at the origin $O(0, 0)$.
2. $f_x(P)$ and $f_y(P)$ at $P(1, 1)$.
3. Compare $|f_x(R)|$ and $|f_y(R)|$ at $R(1, -1)$. Which is greater?

Solution 9.1

1. At $O(0,0)$:
$$f_x(0,0) = 0, f_y(0,0) = -10 < 0$$
2. At $P(1,1)$: As x increases, $f(x,1)$ increases towards 0 from negative values, so $f_x(P) > 0$. At $y = 1$, $f(1,y)$ reaches a local extremum, so $f_y(P) = 0$.
3. At $R(1,-1)$: The contour lines are significantly denser in the y-direction than in the x-direction. Thus:
$$|f_x(R)| > |f_y(R)|$$

9.2 Linear Approximation

Definition 9.2: Linear Function

A **linear function** of two variables is a function of the form

$$f(x,y) = m_x x + m_y y + c$$

The idea of linear approximation is to approximate a function $f(x,y)$ by a linear function $L(x,y)$ that is tangent to the surface defined by f at a given point (a,b).

Given a function $L(x,y) = m_x(x-a) + m_y(y-b) + C$.

Firstly, we want $L(a,b) = f(a,b)$. We then write the linear function as

$$L(x,y) = m_x(x-a) + m_y(y-b) + f(a,b)$$

Secondly, we want the traces of L to be tangent to the traces of f at (a,b). Hence, we have

$$m_x = f_x(a,b), \quad m_y = f_y(a,b)$$

Therefore,

$$L(x,y) = f_x(a,b) \cdot (x-a) + f_y(a,b) \cdot (y-b) + f(a,b)$$

Thus, we now have the following definition of linear approximation:

Definition 9.3: Linear Approximation

The **linear approximation** of a function $f(x, y)$ at the point (a, b) is given by

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b),$$

which is the tangent plane.

Problem 9.2: Finding a Linear Approximation

Find the linear approximation of $f(x, y) = \frac{x^2}{y^2+1}$ at $(4, 1)$. Use your answer to approximate $f(4.01, 0.98)$.

Solution 9.2

First, we find the partial derivatives of $f(x, y) = \frac{x^2}{y^2+1}$:

$$f_x(x, y) = \frac{2x}{y^2 + 1}, \quad f_y(x, y) = -\frac{2x^2y}{(y^2 + 1)^2}$$

At the point $(4, 1)$:

$$f(4, 1) = \frac{16}{2} = 8, \quad f_x(4, 1) = \frac{8}{2} = 4, \quad f_y(4, 1) = -\frac{32}{4} = -8$$

The linear approximation is:

$$z = 8 + 4(x - 4) + (-8)(y - 1)$$

Simplifying:

$$z = 8 + 4x - 16 - 8y + 8 = 4x - 8y$$

To approximate $f(4.01, 0.98)$:

$$f(4.01, 0.98) \approx 4(4.01) - 8(0.98) = 16.04 - 7.84 = 8.2$$

9.3 Differentiability

Definition 9.4: Differentiability

A function $f(x, y)$ is said to be **differentiable** at a point (a, b) if the linear approximation of f at (a, b) is a good approximation of f near (a, b).

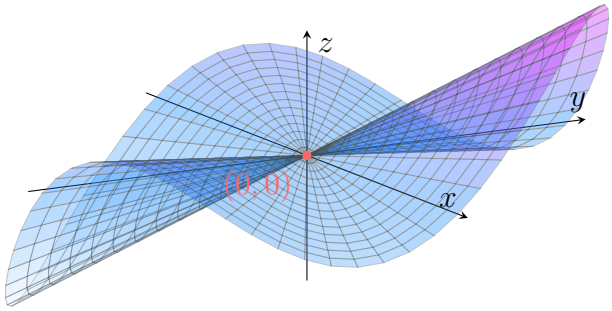
This is equivalent to saying that $f(x, y)$ has a tangent plane at (a, b).

A formal definition of differentiability is as follows:

$$\lim_{(x,y)\to(a,b)} \frac{f(x, y) - L(x, y)}{\sqrt{(x - a)^2 + (y - b)^2}} = 0$$

Example 9.3 (**Non-example:**).

$$g(x, y) = \begin{cases} \dfrac{2xy(x + y)}{x^2 + y^2} & (x, y) \neq (0, 0) \\ \\ 0 & (x, y) = (0, 0) \end{cases}.$$



Note that in single variable calculus, if a function has a derivative at a point, it is also differentiable at that point. However, in multivariable calculus, this is not necessarily the case.

This function $g(x, y)$ is a tricky function. From calculation, we know its partial derivatives,

$\frac{\partial g}{\partial x}$ and $\frac{\partial g}{\partial y}$, exist at $(0,0)$:

$$\begin{aligned} g_x(0,0) &= \lim_{h\to 0} \frac{g(h,0)-g(0,0)}{h} \\ &= \lim_{h\to 0} \frac{0-0}{h} = 0 \\ g_y(0,0) &= \lim_{h\to 0} \frac{g(0,h)-g(0,0)}{h} \\ &= \lim_{h\to 0} \frac{0-0}{h} = 0 \end{aligned}$$

In fact, from all directions, the derivatives of g exists.

However, g is not **differentiable** at $(0,0)$.

To show this, we can use the definition of differentiability.

$$\begin{aligned} \lim_{(x,y)\to(0,0)} \frac{g(x,y)-L(x,y)}{\sqrt{x^2+y^2}} &= \lim_{(x,y)\to(0,0)} \frac{g(x,y)-g(0,0)+g_x(0,0)(x-0)+g_y(0,0)(y-0)}{\sqrt{x^2+y^2}} \\ &= \lim_{(x,y)\to(0,0)} \frac{g(x,y)}{\sqrt{x^2+y^2}} \\ &= \lim_{(x,y)\to(0,0)} \frac{2xy(x+y)}{(x^2+y^2)\sqrt{x^2+y^2}} \\ &= \lim_{(x,y)\to(0,0)} \frac{2xy(x+y)}{(x^2+y^2)^{3/2}} \end{aligned}$$

We approach $(0,0)$ along the line $y = x$:

$$\begin{aligned}
\lim_{(x,y)\to(0,0)} \frac{2xy(x+y)}{(x^2+y^2)^{3/2}} &= \lim_{x\to 0} \frac{2x \cdot x(x+x)}{(x^2+x^2)^{3/2}} \\
&= \lim_{x\to 0} \frac{2x^2(2x)}{(2x^2)^{3/2}} \\
&= \lim_{x\to 0} \frac{4x^3}{(2x^2)^{3/2}} \\
&= \lim_{x\to 0} \frac{4x^3}{2^{3/2}x^3} \\
&= \lim_{x\to 0} \frac{4}{2^{3/2}} \\
&= \frac{4}{2\sqrt{2}} = \sqrt{2} \neq 0
\end{aligned}$$

This shows that linear approximation is not a good approximation of g near $(0,0)$, and hence g is not differentiable at $(0,0)$.

We now introduce a theorem that provides a sufficient condition for differentiability.

Theorem 9.1: Sufficient Condition for Differentiability

If f_x and f_y are continuous on an open disk D, then $f(x,y)$ is differentiable on D.

9.4 Higher-Order Partial Derivatives

Definition 9.5: Higher-Order Partial Derivatives

We denote by f_{xx} the function

$$f_{xx}(x,y) = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial f}{\partial x}$$

and similarly for f_{yy}.

We denote by f_{xy} the function

$$f_{xy}(x,y) = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x \partial y}$$

and similarly for f_{yx}.

Note that the order of differentiation matters for mixed partial derivatives.

The interpretation of higher-order partial derivatives is similar to that of first-order partial derivatives.

- $f_{xx}(a,b)$ is the instantaneous rate of change of f_x at the point (a,b) in the x-direction.
- $f_{xy}(a,b)$ is the instantaneous rate of change of f_x at the point (a,b) in the y-direction.
- $f_{yx}(a,b)$ is the instantaneous rate of change of f_y at the point (a,b) in the x-direction.

Theorem 9.2: Clairaut's Theorem

Suppose that f_{xy} and f_{yx} are continuous on some open disk D. Then

$$f_{xy}(a,b) = f_{yx}(a,b) \quad \text{for all } (a,b) \in D$$

It is similar for functions of more than two variables.

Example 9.4 (Application of Clairaut's Theorem)**.**
Let $g(x,y,z,w) = x^3w^2z^2 + \sin\left(\frac{xy}{z^2}\right)$. Now we want to find g_{zzwx}.

First, we examine the function g. Notice that the first term, $x^3w^2z^2$, is a polynomial function, which is infinitely differentiable over $\mathbb{R}^n$.

The second term, $\sin\left(\frac{xy}{z^2}\right)$, is a composite function. However, sin is infinitely differentiable over $\mathbb{R}$, and $\frac{xy}{z^2}$ also infinitely differentiable over $\mathbb{R}^n, z \neq 0$.

Hence, g is infinitely differentiable over $\mathbb{R}^n, z \neq 0$.

This gives that

$$g_{zzwx} = \frac{\partial g}{\partial z, z, w, x} = g_{wzzx} = g_{xwzz} = \cdots$$

If we calculate this in the original order, we have

$$\begin{aligned} g_{zzwx} &= (((g_z)_z)_w)_x \\ &= (((6x^3w^2z - \frac{2y}{z^3}\cos\left(\frac{xy}{z^2}\right) - \frac{2xy^2}{z^5}\sin\left(\frac{xy}{z^2}\right))_z)_w)_x \\ &= (((6x^3w^2 + \frac{6y}{z^4}\cos\left(\frac{xy}{z^2}\right) + \frac{10xy^2}{z^6}\sin\left(\frac{xy}{z^2}\right) - \frac{4x^2y^3}{z^8}\cos\left(\frac{xy}{z^2}\right))_w)_x \\ &= \cdots \end{aligned}$$

We want to try to avoid this tedious calculation by using Clairaut's Theorem. Rearranging the order of differentiation to g_{xwzz}, we have

$$\begin{aligned} g_{xwzz} &= (((g_x)_w)_z)_z \\ &= (((3x^2w^2z^2 + \frac{y}{z^2}\cos\left(\frac{xy}{z^2}\right))_w)_z)_z \\ &= (((6x^2wz^2 + 0)_z)_z) \\ &= ((12x^2wz + 0)_z) \\ &= 12x^2w \end{aligned}$$

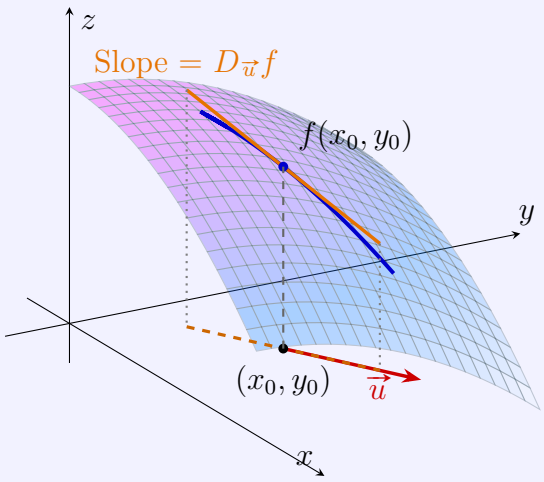
9.5 Directional Derivatives

Definition 9.6: Directional Derivative

Let $\vec{u}$ be a **unit vector**, and let $f(x, y)$ be a function of two variables. The **directional derivative** of f at the point (a, b) in the direction of $\vec{u}$ is defined as

$$D_{\vec{u}}f(\vec{r_0}) = \lim_{t\to 0}\frac{f(\vec{r_0} + t\vec{u}) - f(\vec{r_0})}{t}$$

Here, we visualize the directional derivative as the slope of the tangent line to the curve on the surface defined by f in the direction of $\vec{u}$ at the point (a, b).



Example 9.5 (Directional Derivative of a Function of Two Variables)**.**

Let $f(x, y) = ax + by$, $\vec{u} = \langle \cos\theta, \sin\theta \rangle$. We now want to find $D_{\vec{u}} f$.

$$
\begin{aligned}
D_{\vec{u}} f &= \lim_{t \to 0} \frac{f(\vec{r_0} + t\vec{u}) - f(\vec{r_0})}{t} \\
&= \lim_{t \to 0} \frac{f(x_0 + t\cos\theta, y_0 + t\sin\theta) - f(x_0, y_0)}{t} \\
&= \lim_{t \to 0} \frac{a(x_0 + t\cos\theta) + b(y_0 + t\sin\theta) - (ax_0 + by_0)}{t} \\
&= \lim_{t \to 0} \frac{at\cos\theta + bt\sin\theta}{t} \\
&= a\cos\theta + b\sin\theta \\
&= \vec{u} \cdot \langle a, b \rangle
\end{aligned}
$$

9.6 Gradient

Definition 9.7: Gradient

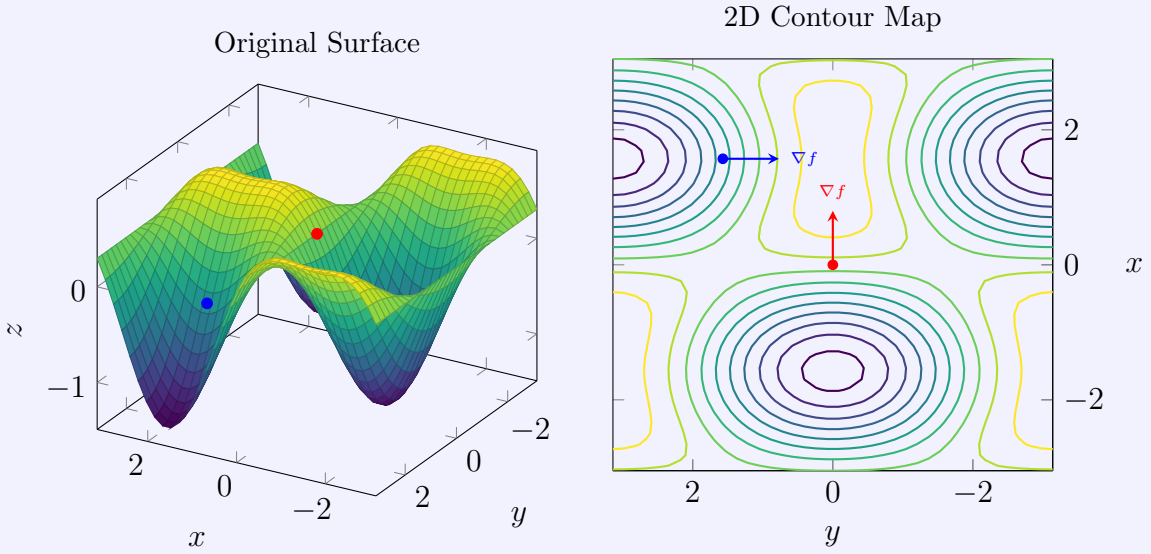
The **gradient** of a function $f(x, y)$ is the vector $\langle a, b \rangle$ such that

$$\langle a, b \rangle = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle$$

We call this vector the **gradient vector** of f and denote it by ∇f.

Here, we visualize the gradient vector as a vector that points in the direc-

tion of the steepest ascent of the surface, as shown below in the graphs.



Theorem 9.3: Gradient Theorem

If f is differentiable at (a, b), then

$$\begin{aligned} D_{\vec{u}} f &= \nabla f \cdot \vec{u} \\ &= \|\nabla f\| \cos\theta \end{aligned}$$

Given this, we know that:

- ∇f points in the direction of the **fastest increase** of f.
- $-\nabla f$ points in the direction of the **fastest decrease** of f.
- ∇f is **perpendicular** to the level curves of f.

Example 9.6 (Gradient of a Function of Two Variables)**.**

Let $f(x, y) = \sin(xy^2)$. We have

$$\nabla f = \langle f_x, f_y \rangle = \left\langle y^2 \cos\left(xy^2\right), 2xy \cos\left(xy^2\right) \right\rangle$$

Given this, we can find the gradient of f at $(0, 1)$.

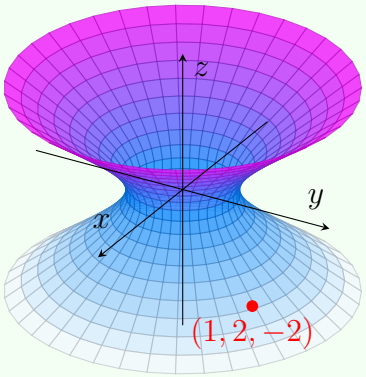
$$\nabla f_{(0,1)} = \langle 1, 0 \rangle$$

Problem 9.3: Finding the Tangent Plane of a Surface

Let $f(x, y, z) = x^2 + y^2 - z^2$, and the surface be defined by $f(x, y, z) = 1$.

Find the tangent plane of the surface at the point $(1, 2, -2)$.

Solution 9.3



First, we compute the gradient of f:

$$\nabla f = \langle f_x, f_y, f_z \rangle = \langle 2x, 2y, -2z \rangle$$

Next, we evaluate the gradient at the point $(1, 2, -2)$:

$$\nabla f(1, 2, -2) = \langle 2, 4, 4 \rangle$$

The equation of the tangent plane at the point (x_0, y_0, z_0) is given by:

$$\nabla f(x_0, y_0, z_0) \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0$$

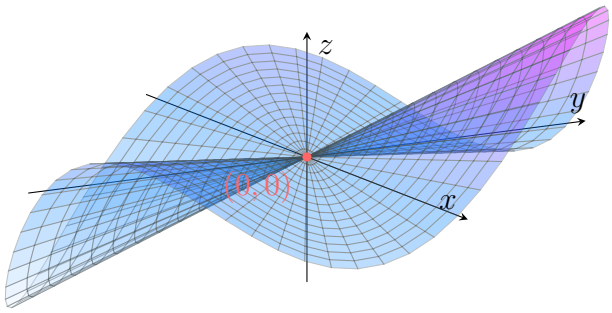
Substituting the values, we have:

$$\begin{aligned}
\langle 2, 4, 4\rangle \cdot \langle x-1, y-2, z+2\rangle &= 0 \\
2(x-1) + 4(y-2) + 4(z+2) &= 0 \\
2x - 2 + 4y - 8 + 4z + 8 &= 0 \\
2x + 4y + 4z - 2 &= 0 \\
x + 2y + 2z - 1 &= 0
\end{aligned}$$

9.7 Differentiability and Continuity

Note that it is dangerous to assume that the existence of partial derivatives implies differentiability.

Recall the example where we proved that the function $g(x, y) = \dfrac{2xy(x+y)}{x^2+y^2}$ is not differentiable at the origin, even though its partial derivatives exist there.



Again, we can compute the gradient of g at $(0,0)$, with $\vec{u} = \langle \cos\theta, \sin\theta \rangle$.

$$\begin{aligned}
D_{\vec{u}} g &= \lim_{t\to 0} \frac{g(\vec{r_0} + t\vec{u}) - g(\vec{r_0})}{t} \\
&= \lim_{t\to 0} \frac{g(t\cos\theta, t\sin\theta) - g(0,0)}{t} \\
&= \lim_{t\to 0} \frac{\dfrac{2(t\cos\theta)(t\sin\theta)(t\cos\theta + t\sin\theta)}{(t\cos\theta)^2 + (t\sin\theta)^2}}{t} \\
&= \lim_{t\to 0} \frac{\dfrac{2t^3(\cos^2\theta\sin\theta + \cos\theta\sin^2\theta)}{t^2(\cos^2\theta + \sin^2\theta)}}{t} \\
&= \lim_{t\to 0} \frac{2t^3(\cos^2\theta\sin\theta + \cos\theta\sin^2\theta)}{t^3} \\
&= \lim_{t\to 0} 2(\cos^2\theta\sin\theta + \cos\theta\sin^2\theta) \\
&= 2(\cos^2\theta\sin\theta + \cos\theta\sin^2\theta) \\
&= 2\cos\theta\sin\theta(\cos\theta + \sin\theta)
\end{aligned}$$

This result is not a constant, and it depends on the direction $\vec{u}$. It yields 0 when $\theta = k\pi/2, k \in \mathbb{Z}$, and ± 1 when $\theta = \pi/4 + k\pi, k \in \mathbb{Z}$.

Note that when

$$D_{\vec{u}} f(x,y) \neq \nabla f \cdot \vec{u}$$

The function is not 'behaving like a plane' at the point (x,y), and hence it is not differentiable at that point.

Theorem 9.4: Differentiability and Continuity

If f_x and f_y are continuous on an open disk D, then f is differentiable on D.

9.8 Chain Rule for Multivariable Functions

Example 9.7 (A Practice Problem)**.**

Given differentiable functions $\vec{r}(t) = \langle x(t), y(t) \rangle$ and $f(x, y)$.

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\vec{r}(t)) = \lim_{h \to 0} \frac{f(\vec{r}(t+h)) - f(\vec{r}(t))}{h}$$

We now compute the linear approximation of f at $\vec{r}(t+h)$:

$$\begin{aligned} f(\vec{r}(t+h)) &\approx f(\vec{r}(t)) + f_x(\vec{r}(t))(x(t+h) - x(t)) + f_y(\vec{r}(t))(y(t+h) - y(t)) \\ &= f(\vec{r}(t)) + f_x(\vec{r}(t))\Delta x + f_y(\vec{r}(t))\Delta y, \\ \text{where } \Delta x &= x(t+h) - x(t), \\ \Delta y &= y(t+h) - y(t) \end{aligned}$$

Hence, the original limit becomes

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} f(\vec{r}(t)) &= \lim_{h \to 0} \frac{f(\vec{r}(t)) + f_x(\vec{r}(t))\Delta x + f_y(\vec{r}(t))\Delta y - f(\vec{r}(t))}{h} \\ &= \lim_{h \to 0} \frac{f_x(\vec{r}(t))\Delta x + f_y(\vec{r}(t))\Delta y}{h} \\ &= \lim_{h \to 0} \left(f_x(\vec{r}(t)) \frac{\Delta x}{h} + f_y(\vec{r}(t)) \frac{\Delta y}{h} \right) \\ &= f_x(\vec{r}(t)) \lim_{h \to 0} \frac{\Delta x}{h} + f_y(\vec{r}(t)) \lim_{h \to 0} \frac{\Delta y}{h} \\ &= f_x(\vec{r}(t)) \frac{\mathrm{d}}{\mathrm{d}t} x(t) + f_y(\vec{r}(t)) \frac{\mathrm{d}}{\mathrm{d}t} y(t) \end{aligned}$$

From this, we can derive the Chain Rule for multivariable functions.

Theorem 9.5: Chain Rule for Two-Variable Functions

Let $f(x, y)$ be a differentiable function of two variables, and let $\vec{r}(t) = \langle x(t), y(t) \rangle$ be

a differentiable vector-valued function of one variable. Then

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}f(\vec{r}(t)) &= f_x(\vec{r}(t))x'(t) + f_y(\vec{r}(t))y'(t)\\ &= \nabla f(\vec{r}(t)) \cdot \vec{r}'(t)\\ &= \frac{\partial f}{\partial x}\frac{\mathrm{d}}{\mathrm{d}t}x + \frac{\partial f}{\partial y}\frac{\mathrm{d}}{\mathrm{d}t}y\end{aligned}$$

If we examine it from basic concepts, we can see how the Chain Rule naturally holds.

$$\begin{aligned}\frac{\Delta f}{\Delta t} &= \frac{\frac{\Delta f}{\Delta x}\Delta x + \frac{\Delta f}{\Delta y}\Delta y}{\Delta t}\\ &= \frac{\Delta f}{\Delta x}\frac{\Delta x}{\Delta t} + \frac{\Delta f}{\Delta y}\frac{\Delta y}{\Delta t}\end{aligned}$$

From the Chain Rule and the definition of directional derivative, we can derive the following theorem.

Theorem 9.6: Directional Derivative and Gradient

Let $f(x, y)$ be a differentiable function of two variables, and let $\vec{u}$ be a unit vector. Then, $\langle u_1, u_2, D_{\vec{u}}f\rangle$ is tangent to f.

Recall that $D_{\vec{u}}f \perp$ the level curves of f. But why is that? Let's explore this concept further.

If $\vec{r}$ follows a level curve of f, then $f(\vec{r}(t))$ is constant. Hence, $\frac{\mathrm{d}}{\mathrm{d}t}f(\vec{r}(t)) = 0$. Meanwhile, $\frac{\mathrm{d}}{\mathrm{d}t}f(\vec{r}(t)) = \nabla f(\vec{r}(t)) \cdot \vec{r}'(t)$.

Therefore, we have

$$\begin{aligned}&\nabla f(\vec{r}(t)) \cdot \vec{r}'(t) = 0\\ \Longrightarrow\ &\nabla f(\vec{r}(t)) \perp \vec{r}'(t)\end{aligned}$$

Problem 9.4: Partial Derivatives of Composite Functions

Given $x = r\cos\theta, y = r\sin\theta, f(x,y) = x^2 + y^2$, find $\dfrac{\partial f}{\partial r}$ and $\dfrac{\partial f}{\partial \theta}$.

Solution 9.4

We can use the Chain Rule to find the partial derivatives of f with respect to r and θ.

$$\begin{aligned}
\frac{\partial f}{\partial r} &= \frac{\partial f}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial r} \\
&= 2x\cos\theta + 2y\sin\theta \\
&= 2(r\cos\theta)\cos\theta + 2(r\sin\theta)\sin\theta \\
&= 2r(\cos^2\theta + \sin^2\theta) \\
&= 2r
\end{aligned}$$

Similarly, we can find $\frac{\partial f}{\partial \theta}$:

$$\begin{aligned}
\frac{\partial f}{\partial \theta} &= \frac{\partial f}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial \theta} \\
&= 2x(-r\sin\theta) - 2y(r\cos\theta) \\
&= -2rx\sin\theta - 2ry\cos\theta \\
&= -2r(r\cos\theta)\sin\theta - 2r(r\sin\theta)\cos\theta \\
&= -2r^2\cos\theta\sin\theta - 2r^2\sin\theta\cos\theta \\
&= -4r^2\cos\theta\sin\theta
\end{aligned}$$

Hereby, we have found the partial derivative for f without directly substituting x and y in terms of r and θ.

Theorem 9.7: Chain Rule for Multivariable Functions

Let $f(x_1, x_2, \ldots, x_n)$ be a differentiable function of n variables, and let $x_i =$

$x_i(t_1, t_2, \ldots, t_m)$ for $i = 1, 2, \ldots, n$. Then,

$$\begin{aligned}\frac{\partial f}{\partial t_j} &= \frac{\partial f}{\partial x_1}\frac{\partial x_1}{\partial t_j} + \frac{\partial f}{\partial x_2}\frac{\partial x_2}{\partial t_j} + \cdots + \frac{\partial f}{\partial x_n}\frac{\partial x_n}{\partial t_j} \\ &= \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}\frac{\partial x_i}{\partial t_j}, \quad j = 1, 2, \ldots, m.\end{aligned}$$

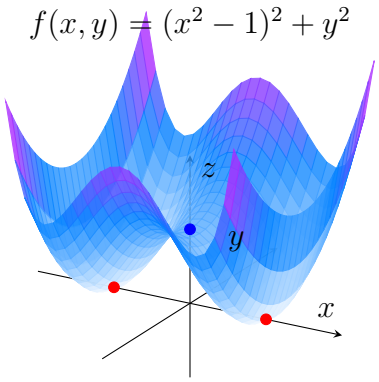
10 Optimization and Lagrange Multipliers

10.1 Extrema of Multivariable Functions

Definition 10.1: Local Maximum and Minimum

For a function $f(x, y)$ and a point (a, b) in its domain, we say that (a, b) is a

- **local maximum** if $f(a, b) \geq f(x, y),\ \forall (x, y)$ in a neighborhood of (a, b).
- **local minimum** if $f(a, b) \leq f(x, y),\ \forall (x, y)$ in a neighborhood of (a, b).



Definition 10.2: Critical Point

We call a point $P(a, b)$ a **critical point** of f if

$$\nabla f(P) = \vec{0}$$

Theorem 10.1: Critical Points and Local Extrema

If $P(a, b)$ is a local extremum of f, then P is a critical point of f.

For single variable functions, we can use the second derivative test to determine the nature of a point on the function. For $f(x)$, if $f'(a) = 0$, then:

- If $f''(a) > 0$, then f has a local minimum at $x = a$.
- If $f''(a) < 0$, then f has a local maximum at $x = a$.
- If $f''(a) = 0$, the test is inconclusive.

For multivariable functions, we can use its quadratic approximation to determine the nature

of a critical point.

Problem 10.1: Determining the Critical Points of a Function

Find critical points of the function $f(x, y) = x^4 + y^4 + 4xy$. Which are local maxima, local minima, and saddle points?

Solution 10.1

First, we find the critical points by setting the gradient of f to zero. The gradient is given by:

$$\nabla f(x, y) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle = (4x^3 + 4y, 4y^3 + 4x)$$

Setting each component to zero gives us the system of equations:

$$4x^3 + 4y = 0$$
$$4y^3 + 4x = 0$$

This implies $y = -x^3$. Substituting $y = -x^3$ into the second equation gives:

$$4(-x^3)^3 + 4x = 0 \implies -4x^9 + 4x = 0 \implies 4x(-x^8 + 1) = 0$$

This gives us $x = 0$ or $x^8 = 1$. The equation $x^8 = 1$ has solutions $x = \pm 1$. So the critical points are $(0, 0)$, $(1, -1)$, and $(-1, 1)$.

To determine the nature of these critical points, we write $f(x, y)$ in the quadratic approximation form around each critical point.

$$f(x, y) \approx a + bx + cy + Ax^2 + Bxy + Cy^2$$
$$\text{If } \nabla f(0, 0) = 0, \text{ then } b = c = 0, a = f(0, 0)$$
$$\implies f(x, y) \approx f(0, 0) + Ax^2 + Bxy + Cy^2$$

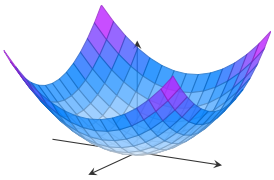
The second derivative is then given by

$$\frac{\partial^2 f}{\partial x \partial x} = 2A$$
$$\frac{\partial^2 f}{\partial y \partial y} = 2C$$
$$\frac{\partial^2 f}{\partial x \partial y} = B$$

Fact: If (a, b) is a critical point of f, then the quadratic approximation is given by

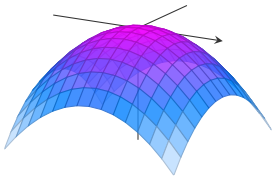
$$Q(x, y) = \frac{1}{2} f_{xx}(a, b)(x - a)^2 + f_{xy}(a, b)(x - a)(y - b) + \frac{1}{2} f_{yy}(a, b)(y - b)^2$$

There are three possible nondegenerate behaviors for a quadratic function near a critical point:



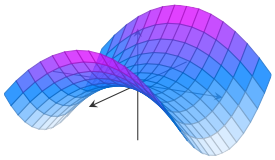
$Q(x, y) = x^2 + y^2$

Minimum



$Q(x, y) = -x^2 - y^2$

Maximum



$Q(x, y) = x^2 - y^2$

Saddle

For a general quadratic function

$$Q(x, y) = Ax^2 + Bxy + Cy^2,$$

its behavior is determined by $4AC - B^2$. For the quadratic approximation of f at a critical point (a, b),

$$A = \frac{1}{2} f_{xx}(a, b), \qquad B = f_{xy}(a, b), \qquad C = \frac{1}{2} f_{yy}(a, b).$$

Hence,

$$4AC - B^2 = f_{xx}(a, b) f_{yy}(a, b) - f_{xy}(a, b)^2.$$

Definition 10.3: Second Derivative Test

Let (a, b) be a critical point of f, and define

$$D = f_{xx}(a, b)f_{yy}(a, b) - f_{xy}(a, b)^2.$$

Then:

- If $D > 0$ and $f_{xx}(a, b) > 0$, then (a, b) is a **local minimum**.
- If $D > 0$ and $f_{xx}(a, b) < 0$, then (a, b) is a **local maximum**.
- If $D < 0$, then (a, b) is a **saddle point**.
- If $D = 0$, the test is **inconclusive**.

Example 10.1 (Usage of the Discriminant)**.**

We have $f(x, y) = x^2 + y^2 - 3xy$.

From calculation, we know that $\nabla f = \langle 2x - 3y, 2y - 3x\rangle$. Setting this equal to zero gives us the system of equations:

$$2x - 3y = 0 \quad \text{and} \quad 2y - 3x = 0.$$

Solving this system, we find that the only critical point is $(0, 0)$.

The discriminant is given by $D = (2)(2) - (-3)^2 = 4 - 9 = -5 < 0$.

We now know that $(0, 0)$ is a saddle.

Definition 10.4: Global Extrema

Given a function $f(x, y)$ defined on some domain $\mathcal{D}$, we say that $(a, b) \in \mathcal{D}$ is a

- **global maximum** if $f(a, b) \geq f(x, y),\ \forall (x, y) \in \mathcal{D}$.
- **global minimum** if $f(a, b) \leq f(x, y),\ \forall (x, y) \in \mathcal{D}$.

10.2 Domains and Boundaries

Definition 10.5: Interior and Boundary Points

Let $\mathcal{D} \subseteq \mathbb{R}^2$ be a domain.

- A point $(a, b) \in \mathcal{D}$ is an **interior point** if is has some disk around it in $\mathcal{D}$.
- A point $(a, b) \in \mathcal{D}$ is a **boundary point** such that every disk around it intersects $\mathcal{D}$ and $\mathbb{R}^2 \setminus \mathcal{D}$.

* $\mathbb{R}^2 \setminus D$ means the set of points in $\mathbb{R}^2$ that are not in D.

Definition 10.6: Closed and Bounded Sets

A set $\mathcal{D} \subseteq \mathbb{R}^2$ is **closed** if it contains all of its boundary points, and **bounded** if it can be contained in a disk of finite radius.

Theorem 10.2: Extreme Value Theorem

Let f be a continuous function on a closed and bounded set $\mathcal{D} \subseteq \mathbb{R}^2$.

Then f attains both a global maximum and a global minimum on $\mathcal{D}$.

Example 10.2 (Finding Global Extrema)**.**

Given the function $f(x, y) = x^2 + y^2 + xy$, and the domain $D = \{(x, y) : x^2 + y^2 \leq 4\}$.

We set the gradient of f to zero to find critical points:

$$\nabla f = \langle 2x + y, 2y + x \rangle = \langle 0, 0 \rangle .$$

This gives us the system of equations:

$$2x + y = 0 \quad \text{and} \quad 2y + x = 0.$$

Solving this system, we find that the only critical point is $(0, 0)$.

Since $\mathcal{D}$ is closed and bounded, and f is continuous on $\mathcal{D}$, the Extreme Value Theorem guarantees that f attains both a global maximum and a global minimum on $\mathcal{D}$. Therefore, we need to check the boundary of $\mathcal{D}$ to find the global extrema.

The boundary of $\mathcal{D}$ is the circle defined by $x^2 + y^2 = 4$. We can parameterize this boundary

using polar coordinates:

$$x = 2\cos(t), \quad y = 2\sin(t), \quad 0 \leq t < 2\pi.$$

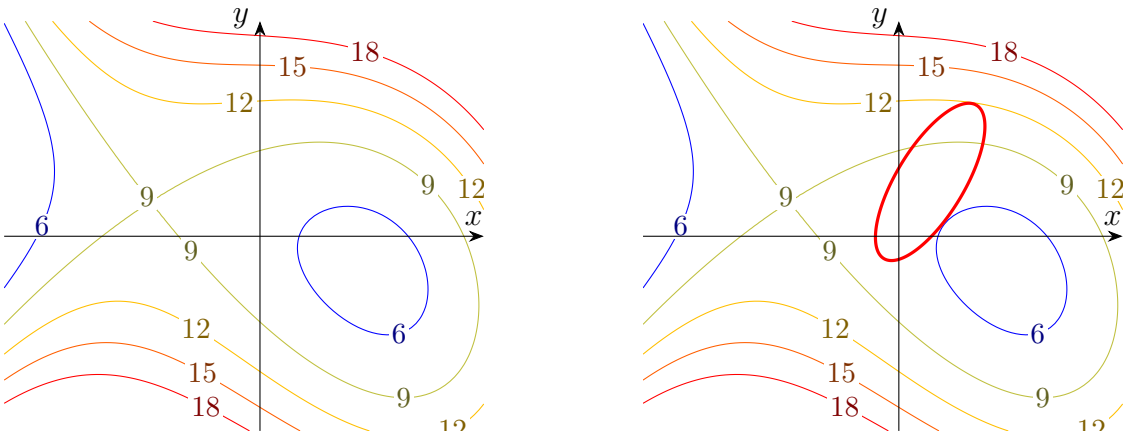
Hence,

$$\begin{aligned} f(2\cos(t), 2\sin(t)) &= (2\cos(t))^2 + (2\sin(t))^2 + (2\cos(t))(2\sin(t)) \\ &= 4\cos^2(t) + 4\sin^2(t) + 4\cos(t)\sin(t) \\ &= 4 + 4\cos(t)\sin(t) \\ &= 4 + 2\sin(2t). \end{aligned}$$

Therefore, the maximum value of f on the boundary is $4 + 2 = 6$ at $x = 2, y = 0$, and the minimum value is $4 - 2 = 2$ at $x = -2, y = 0$.

10.3 Lagrange Multipliers

Example 10.3 (Extrema of a Function on a Curve)**.**
Below is a function f and a curve C given by $g(x, y) = c$.



In this example, the extrema under constraints occur where the level curves of f are tangent to the constraint curve C. This is a particularly lucky situation, as we can visually identify the points of tangency.

Theorem 10.3: Lagrange Multipliers

Let $f(x, y)$ be a function and $g(x, y) = c$ be a constraint curve. If $g(x, y) = c$ is tangent to a level curve of f, then

$$\nabla f \parallel \nabla g \implies \nabla f = \lambda \nabla g$$

Therefore, to find the extrema of f under the constraint $g(x, y) = c$, we can solve the system of equations:

$$\nabla f = \lambda \nabla g, \quad g(x, y) = c$$

for x, y, and λ, assuming that $\nabla g \neq 0$ at the points of interest.

Example 10.4 (Using Lagrange Multipliers)**.**

We want to find the extrema of $f(x, y) = x + 2y$ subject to the constraint $3x^2 + 2y^2 = 7$.

First, we compute the gradients:

$$\nabla f = \langle 1, 2 \rangle, \quad \nabla g = \langle 6x, 4y \rangle.$$

Setting $\nabla f = \lambda \nabla g$ gives us the system of equations:

$$\begin{cases} 1 = \lambda(6x) \\ 2 = \lambda(4y) \\ 3x^2 + 2y^2 = 7 \end{cases}$$

Solving the first two equations for λ, we get:

$$\begin{cases} \lambda = \dfrac{1}{6x} \\ \lambda = \dfrac{1}{2y} \end{cases}$$

Setting these equal gives us:

$$\frac{1}{6x} = \frac{1}{2y} \implies 2y = 6x \implies y = 3x.$$

Substituting $y = 3x$ into the constraint equation gives:

$$\begin{aligned} 3x^2 + 2(3x)^2 &= 7 \\ 3x^2 + 18x^2 &= 7 \\ 21x^2 &= 7 \\ x^2 &= \frac{1}{3} \\ x &= \pm\frac{1}{\sqrt{3}}. \end{aligned}$$

Thus, the corresponding y values are:

$$y = 3\left(\pm\frac{1}{\sqrt{3}}\right) = \pm\sqrt{3}.$$

Therefore, the points of interest are:

$$\left(\frac{1}{\sqrt{3}}, \sqrt{3}\right) \text{ and } \left(-\frac{1}{\sqrt{3}}, -\sqrt{3}\right).$$

Example 10.5 (Another Example of Lagrange Multipliers)**.**
Given a function $Q(K, L) = AK^{\alpha}L^{1-\alpha}$. This describes the output when K units of capital and L units of labor are used, where $A > 0$ is a constant and $0 < \alpha < 1$.

Suppose that $A = 1$ and $\alpha = \frac{3}{5}$. Given a budget of \$12 with the cost of capital being \$3 per unit and the cost of labor being \$4 per unit, we want to maximize $Q(K, L)$ subject to the constraint $2K + 3L = 12$.

The process is similar to the previous example. We first rewrite the constraint as $g(K, L) = 2K + 3L - 12 = 0$. Then, we compute the gradients:

$$\begin{aligned} \nabla Q &= \left\langle \frac{3}{5}K^{-2/5}L^{2/5}, \frac{2}{5}K^{3/5}L^{-3/5} \right\rangle \\ \nabla g &= \langle 2, 3 \rangle. \end{aligned}$$

Setting $\nabla Q = \lambda \nabla g$ gives us the system of equations:

$$\begin{cases} \frac{3}{5}K^{-2/5}L^{2/5} = 2\lambda \\ \frac{2}{5}K^{3/5}L^{-3/5} = 3\lambda \\ 2K + 3L = 12 \end{cases}$$

Solving the first two equations for λ, we get:

$$\frac{3}{10}K^{-2/5}L^{2/5} = \frac{2}{15}K^{3/5}L^{-3/5} \implies 9L = 4K \implies L = \frac{4}{9}K.$$

Substituting $L = \frac{4}{9}K$ into the constraint equation gives:

$$\begin{aligned} 2K + 3\left(\frac{4}{9}K\right) &= 12 \\ 2K + \frac{4}{3}K &= 12 \\ \frac{10}{3}K &= 12 \\ K &= \frac{36}{10} = 3.6. \end{aligned}$$

Then, substituting $K = 3.6$ into $L = \frac{4}{9}K$ gives:

$$L = \frac{4}{9} \cdot 3.6 = 1.6.$$

Therefore, the optimal allocation of resources is $K = 3.6$ units of capital and $L = 1.6$ units of labor.

11 Double Integrals

Definition 11.1: Imprecise Definition of Double Integral

Let $f(x, y)$ be a function defined on a closed and bounded region $\mathcal{D}$ in the xy-plane. The **double integral** is defined by

$$\iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A$$

where $\mathrm{d}A$ is the area element in the xy-plane.

This integral can be interpreted as the volume under the surface $z = f(x, y)$ and above the region $\mathcal{D}$ in the xy-plane.

Below are some properties of double integrals that will be useful in evaluating them.

- $\iint_{\mathcal{D}} f(x, y) + g(x, y)\,\mathrm{d}A = \iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A + \iint_{\mathcal{D}} g(x, y)\,\mathrm{d}A$
- $\iint_{\mathcal{D}} c \cdot f(x, y)\,\mathrm{d}A = c \cdot \iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A$
- If $f(x, y) \leq g(x, y)$ for all $(x, y) \in \mathcal{D}$, then $\iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A \leq \iint_{\mathcal{D}} g(x, y)\,\mathrm{d}A$.

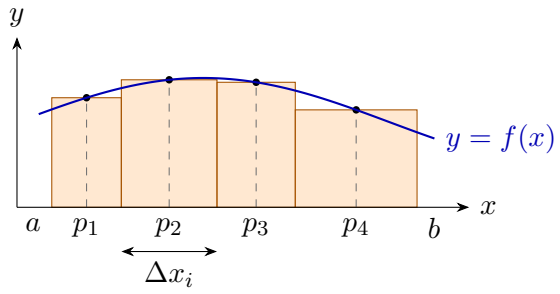
11.1 Riemann Sums and the Definition of Double Integrals

Recall the concept of Riemann sums for single-variable functions.

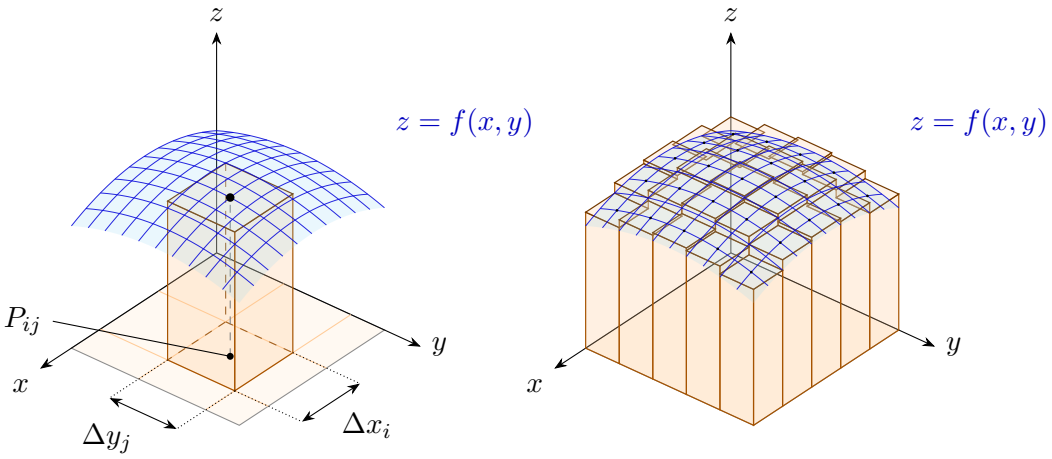
The Riemann sum for a function $f(x)$ over an interval $[a, b]$ is given by

$$\sum_{i=1}^{n} f(p_i)\Delta x_i$$

where Δx_i is the width of the i-th subinterval, and p_i is a sample point in that subinterval, as indicated in the figure below.



We can extend this idea to functions of two variables. Assume that $\mathcal{D} = [a, b] \times [c, d]$ is rectangular region in the xy-plane.



By this, we could write a precise definition of the double integral as follows:

Definition 11.2: Precise Definition of Double Integral

Let $f(x, y)$ be a function defined on a closed and bounded region $\mathcal{D}$ in the xy-plane.

We partition $\mathcal{D}$ into m subintervals in the x-direction and n subintervals in the y-direction, forming a grid of rectangles.

Let Δx_i and Δy_j be the widths of the subintervals in the x and y directions, respectively, and let P_{ij} be a sample point in the rectangle formed by the i-th subinterval in the x-direction and the j-th subinterval in the y-direction.

The **double integral** of $f(x, y)$ over the region $\mathcal{D}$ is defined as the limit of the Riemann

sum as the maximum width of the subintervals approaches zero:

$$\iint_{\mathcal{D}} f(x,y)\,\mathrm{d}A = \lim_{\|P\|\to 0} \sum_{i=1}^{m}\sum_{j=1}^{n} f(P_{ij})\Delta A_{ij}$$

Notice that the ΔA_{ij} here does not always exist. If it does, we call the function $f(x,y)$ **integrable** over the region $\mathcal{D}$. Also, if a function is continuous over a closed and bounded region, then it is integrable over that region.

11.2 Iterated Integrals and Fubini's Theorem

To evaluate double integrals, we can use Fubini's Theorem, which allows us to compute the double integral as an iterated integral.

Theorem 11.1: Fubini's Theorem

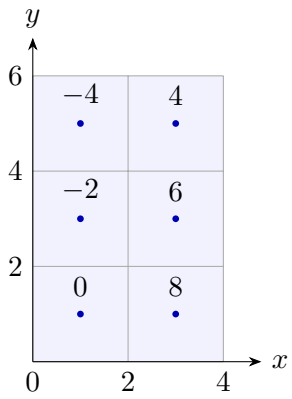
$$\iint_{\mathcal{D}} f(x,y)\,\mathrm{d}A = \int_a^b \int_c^d f(x,y)\,\mathrm{d}y\,\mathrm{d}x = \int_c^d \int_a^b f(x,y)\,\mathrm{d}x\,\mathrm{d}y$$

Using this theorem, we can evaluate a double integral by computing two single-variable integrals in succession.

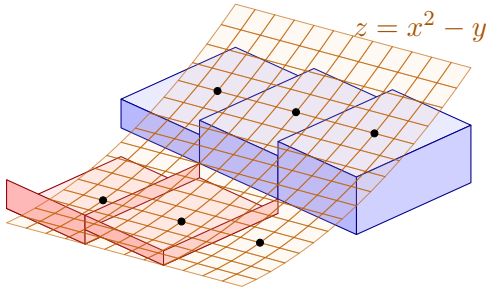
Example 11.1 (Evaluating a Double Integral)**.**

Let $f(x,y) = x^2 - y$ and $\mathcal{D} = [0,4] \times [0,6]$. We first approximate $\iint_{\mathcal{D}} f(x,y)\,\mathrm{d}A$ using a midpoint Riemann sum. Divide $[0,4]$ into two equal intervals and $[0,6]$ into three equal intervals. This gives six rectangles, each with area $\Delta A = 2 \cdot 2 = 4$. The sample points are their centers:

$$(1,1),\ (1,3),\ (1,5),\ (3,1),\ (3,3),\ (3,5).$$



Midpoint samples, labeled by $f(x_i, y_i)$.



Midpoint prisms and the surface $z = x^2 - y$.

Thus the midpoint Riemann sum is

$$\begin{aligned} S &= \sum_{i=1}^{6} f(x_i, y_i)\Delta A \\ &= 4\big(0 - 2 - 4 + 8 + 6 + 4\big) = 48. \end{aligned}$$

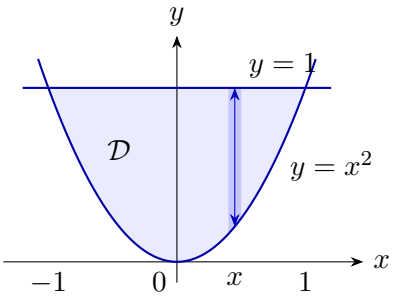
The negative terms represent contributions below the xy-plane.

By Fubini's theorem, the exact value is

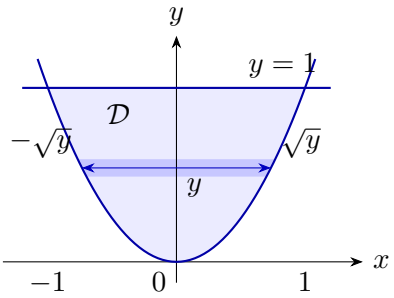
$$\begin{aligned} \iint_{\mathcal{D}} f(x,y)\,\mathrm{d}A &= \int_0^4 \int_0^6 (x^2 - y)\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_0^4 \left[x^2 y - \frac{y^2}{2}\right]_0^6 \mathrm{d}x \\ &= \int_0^4 (6x^2 - 18)\,\mathrm{d}x \\ &= \big[2x^3 - 18x\big]_0^4 = \boxed{56}. \end{aligned}$$

For a non-rectangular region, the limits of the inner integral may depend on the outer variable.

Example 11.2 (Setting Up a Double Integral over a Non-Rectangular Region)**.**
Let $\mathcal{D}$ be the region bounded by $y = x^2$ and $y = 1$, and let f be continuous on $\mathcal{D}$. The curves intersect where $x^2 = 1$, so $x = \pm 1$.



Vertical slice: $x^2 \le y \le 1$.



Horizontal slice: $-\sqrt{y} \le x \le \sqrt{y}$.

For vertical slices, fix x and let y run from the lower boundary $y = x^2$ to the upper boundary $y = 1$. The slices cover the region as x runs from -1 to 1, so

$$\mathcal{D} = \{(x, y) \mid -1 \le x \le 1,\ x^2 \le y \le 1\}.$$

Hence

$$\iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A = \int_{-1}^{1} \int_{x^2}^{1} f(x, y)\,\mathrm{d}y\,\mathrm{d}x.$$

For horizontal slices, fix y and solve $x^2 \le y$ to obtain $-\sqrt{y} \le x \le \sqrt{y}$. The slices cover the region as y runs from 0 to 1, so

$$\mathcal{D} = \{(x, y) \mid 0 \le y \le 1,\ -\sqrt{y} \le x \le \sqrt{y}\}.$$

Thus the other order of integration is

$$\iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A = \int_{0}^{1} \int_{-\sqrt{y}}^{\sqrt{y}} f(x, y)\,\mathrm{d}x\,\mathrm{d}y.$$

11.3 Simple Regions

Definition 11.3: Simple Region

A region $\mathcal{D}$ is

- **Vertically simple** if

$$\mathcal{D} = \{(x, y) \mid a \le x \le b,\ g_1(x) \le y \le g_2(x)\},$$

where g_1 and g_2 are continuous on $[a, b]$ and $g_1(x) \leq g_2(x)$. Each vertical line intersects $\mathcal{D}$ in a single line segment, a point, or not at all.

- **Horizontally simple** if

$$\mathcal{D} = \{(x, y) \mid c \leq y \leq d,\ h_1(y) \leq x \leq h_2(y)\},$$

where h_1 and h_2 are continuous on $[c, d]$ and $h_1(y) \leq h_2(y)$. Each horizontal line intersects $\mathcal{D}$ in a single line segment, a point, or not at all.

If f is continuous on the corresponding simple region, then

$$\text{Vertically simple:} \quad \iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\,\mathrm{d}y\,\mathrm{d}x,$$

$$\text{Horizontally simple:} \quad \iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y)\,\mathrm{d}x\,\mathrm{d}y.$$

11.4 Average Value of a Function

Definition 11.4: Average Value of a Function

Let f be continuous on a closed and bounded region $\mathcal{D}$ in the xy-plane. The **average value** of f over $\mathcal{D}$ is defined by

$$\bar{f} = \frac{1}{A(\mathcal{D})} \iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A,$$

where $A(\mathcal{D})$ is the area of the region $\mathcal{D}$.

Problem 11.1: Average Value of a Function

Let $f(x, y) = x^2 + y^2$ and $\mathcal{D}$ be the region bounded by $y = x^2$ and $y = 1$. Find the average value of f over $\mathcal{D}$.

Solution 11.1

By definition, the average value is

$$\bar{f} = \frac{1}{A(\mathcal{D})} \iint_{\mathcal{D}} f(x,y)\,\mathrm{d}A.$$

The area of $\mathcal{D}$ is

$$\begin{aligned} A(\mathcal{D}) &= \int_{-1}^{1}\int_{x^2}^{1} 1\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_{-1}^{1}(1-x^2)\,\mathrm{d}x = \left[x - \frac{x^3}{3}\right]_{-1}^{1} \\ &= \frac{4}{3}. \end{aligned}$$

Next, we compute the double integral:

$$\begin{aligned} \iint_{\mathcal{D}} f(x,y)\,\mathrm{d}A &= \int_{-1}^{1}\int_{x^2}^{1}(x^2+y^2)\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_{-1}^{1}\left[x^2y + \frac{y^3}{3}\right]_{y=x^2}^{y=1}\mathrm{d}x \\ &= \int_{-1}^{1}\left(x^2 + \frac{1}{3} - x^4 - \frac{x^6}{3}\right)\mathrm{d}x \\ &= \left[\frac{x^3}{3} + \frac{x}{3} - \frac{x^5}{5} - \frac{x^7}{21}\right]_{-1}^{1} \\ &= 2\left(\frac{1}{3} + \frac{1}{3} - \frac{1}{5} - \frac{1}{21}\right) = 2\left(\frac{14+14-21-5}{105}\right) = 2\left(\frac{2}{15}\right) = \frac{4}{15}. \end{aligned}$$

Thus the average value is

$$\frac{1}{\frac{4}{3}} \cdot \frac{4}{15} = \frac{1}{5}.$$

Theorem 11.2: Linearity of Double Integrals

Let f be a continuous function on the xy-plane. If $\mathcal{D} = \mathcal{D}_1 \cup \mathcal{D}_2$, where $\mathcal{D}_1$ and $\mathcal{D}_2$

overlap only on their boundaries or not at all, then

$$\iint_{\mathcal{D}} f(x,y)\,\mathrm{d}A = \iint_{\mathcal{D}_1} f(x,y)\,\mathrm{d}A + \iint_{\mathcal{D}_2} f(x,y)\,\mathrm{d}A.$$

By this theorem, we can compute the double integral over a complicated region by breaking it into simpler regions.

12 Triple Integrals

12.1 Interpretations of Triple Integrals

Let $\mathcal{W} \subseteq \mathbb{R}^3$ be a solid region and let $f(x, y, z)$ be integrable on $\mathcal{W}$. The triple integral

$$\iiint_{\mathcal{W}} f(x, y, z)\, \mathrm{d}V$$

has several important interpretations:

1. If f is a density function with units of amount per unit volume, then

   $$\iiint_{\mathcal{W}} f(x, y, z)\, \mathrm{d}V$$

   is the total amount contained in $\mathcal{W}$. In particular, if $f = \rho$ is a mass density, then the integral gives the mass of the solid.

2. Taking $f = 1$ gives the volume of the region:

   $$\mathrm{Vol}(\mathcal{W}) = \iiint_{\mathcal{W}} 1\, \mathrm{d}V.$$

3. The average value of f over $\mathcal{W}$ is

   $$f_{\mathrm{avg}} = \frac{1}{\mathrm{Vol}(\mathcal{W})} \iiint_{\mathcal{W}} f(x, y, z)\, \mathrm{d}V.$$

4. Geometrically,

   $$\iiint_{\mathcal{W}} f(x, y, z)\, \mathrm{d}V$$

   can be viewed as the signed hypervolume of the four-dimensional region determined by the graph of f over $\mathcal{W}$.

12.2 Evaluating Triple Integrals

As with double integrals, a triple integral may be approximated by a Riemann sum or evaluated exactly by writing it as an iterated integral.

Theorem 12.1: Fubini's Theorem for Triple Integrals

If f is continuous on the box

$$\mathcal{B} = [a, b] \times [c, d] \times [p, q],$$

then

$$\iiint_{\mathcal{B}} f(x, y, z)\, \mathrm{d}V = \int_a^b \int_c^d \int_p^q f(x, y, z)\, \mathrm{d}z\, \mathrm{d}y\, \mathrm{d}x.$$

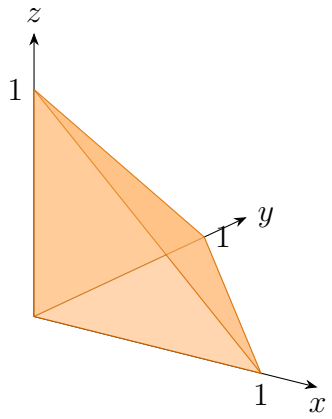
The same conclusion holds for any of the other five orders of integration.

More generally, suppose $\mathcal{W}$ lies over a domain $\mathcal{D}$ in the xy-plane and is bounded below and above by $z = z_1(x, y)$ and $z = z_2(x, y)$, respectively. Then

$$\iiint_{\mathcal{W}} f(x, y, z)\, \mathrm{d}V = \iint_{\mathcal{D}} \left(\int_{z_1(x,y)}^{z_2(x,y)} f(x, y, z)\, \mathrm{d}z \right) \mathrm{d}A.$$

Example 12.1 (Volume of a Tetrahedron)**.**

The plane $x + y + z = 1$ cuts off a tetrahedron $\mathcal{W}$ in the first octant. Describe $\mathcal{W}$ using inequalities in rectangular coordinates, then find its volume.



Solution 12.1

Because $\mathcal{W}$ lies in the first octant and below the plane $z = 1 - x - y$,

$$\mathcal{W} = \{(x, y, z) : x \geq 0,\ y \geq 0,\ z \geq 0,\ x + y + z \leq 1\}.$$

Using the order $\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x$, the bounds are

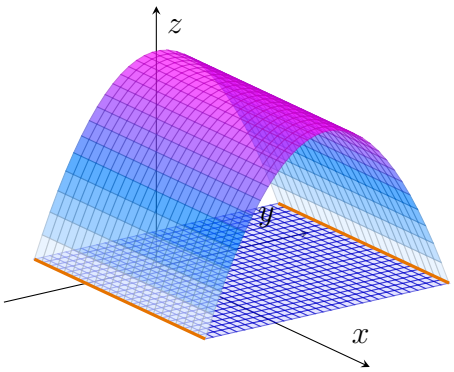
$$0 \leq x \leq 1, \qquad 0 \leq y \leq 1 - x, \qquad 0 \leq z \leq 1 - x - y.$$

Therefore,

$$\begin{aligned}
\mathrm{Vol}(\mathcal{W}) &= \int_0^1 \int_0^{1-x} \int_0^{1-x-y} 1 \,\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x \\
&= \int_0^1 \int_0^{1-x} (1 - x - y)\,\mathrm{d}y\,\mathrm{d}x \\
&= \frac{1}{2}\int_0^1 (1 - x)^2\,\mathrm{d}x = \frac{1}{6}.
\end{aligned}$$

Example 12.2 (Mass Between Two Surfaces)**.**

Let $\mathcal{W}$ be the solid bounded by $z = 1$, $z = 2 - y^2$, $x = 0$, and $x = 1$. The density at (x, y, z) is $f(x, y, z)$ kilograms per cubic meter. Write an iterated integral for the mass of $\mathcal{W}$.



Solution 12.2

The two z-surfaces intersect when

$$1 = 2 - y^2 \iff y = \pm 1.$$

Thus

$$\mathcal{W} = \{(x, y, z) : 0 \le x \le 1,\ -1 \le y \le 1,\ 1 \le z \le 2 - y^2\}.$$

The mass is

$$M = \int_0^1 \int_{-1}^1 \int_1^{2-y^2} f(x, y, z)\, \mathrm{d}z\, \mathrm{d}y\, \mathrm{d}x.$$

Example 12.3 (Choosing an Order of Integration)**.**

Let $\mathcal{W}$ be the solid in the first octant bounded by

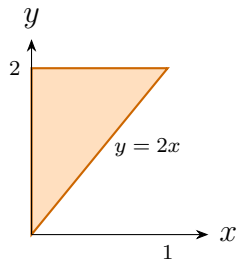
$$z = 4 - y^2, \qquad y = 2x, \qquad z = 0, \qquad x = 0.$$

Calculate

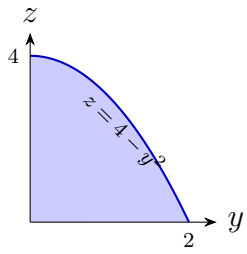
$$\iiint_{\mathcal{W}} xyz\, \mathrm{d}V.$$

The three coordinate-plane projections of $\mathcal{W}$ are shown below.

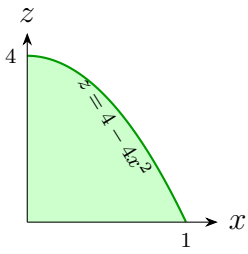
Projection onto the xy-plane



Projection onto the yz-plane



Projection onto the xz-plane



Solution 12.3

The projection onto the xy-plane is

$$D_{xy} = \{(x, y) : 0 \le y \le 2,\ 0 \le x \le \tfrac{y}{2}\}.$$

For each $(x, y) \in D_{xy}$, the variable z ranges from the xy-plane to the parabolic cylinder:

$$0 \le z \le 4 - y^2.$$

Therefore,

$$\begin{aligned}\iiint_{\mathcal{W}} xyz \,\mathrm{d}V &= \int_0^2 \int_0^{y/2} \int_0^{4-y^2} xyz \,\mathrm{d}z \,\mathrm{d}x \,\mathrm{d}y \\ &= \int_0^2 \int_0^{y/2} \frac{xy}{2}(4-y^2)^2 \,\mathrm{d}x \,\mathrm{d}y \\ &= \frac{1}{16}\int_0^2 y^3(4-y^2)^2 \,\mathrm{d}y.\end{aligned}$$

With $u = y^2$, so that $y^3 \,\mathrm{d}y = \frac{u}{2}\,\mathrm{d}u$,

$$\begin{aligned}\iiint_{\mathcal{W}} xyz \,\mathrm{d}V &= \frac{1}{32}\int_0^4 u(4-u)^2 \,\mathrm{d}u \\ &= \frac{2}{3}.\end{aligned}$$